

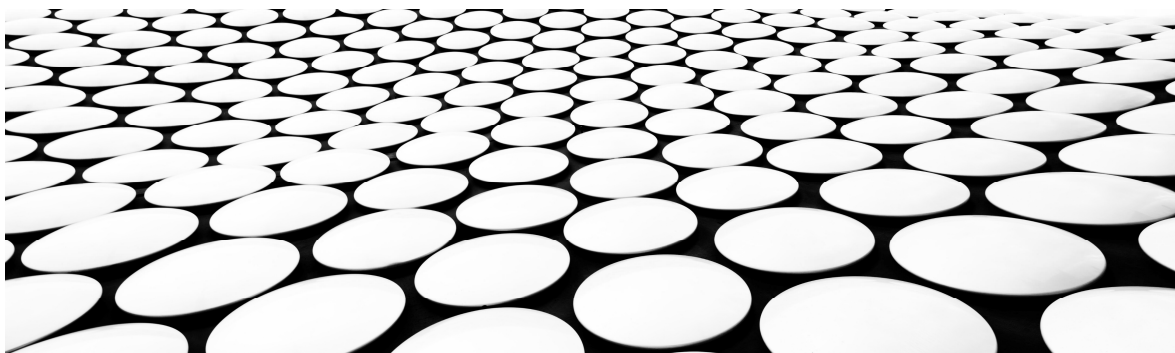


ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN
KHOA ĐIỆN TỬ - VIỄN THÔNG

THIẾT KẾ VI MẠCH ĐIỆN TỬ

CHAPTER 2: CMOS DEVICES STRUCTURES

LÊ ĐỨC HÙNG



CONTENT

- PN Junctions
- MOS Transistors
- Capacitor
- Resistor
- Inductor

PN JUNCTIONS

How are PN Junctions used in CMOS?

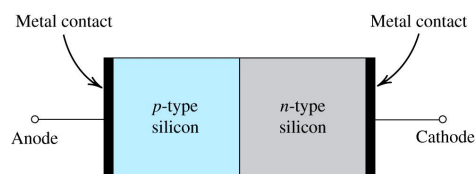
- A very important role in all semiconductor devices
- PN junctions are used to electrically isolate one semiconductor region from another
- Can be used as PN diodes
- ESD protection
- Creation of the thermal voltage for bandgap purposes
- Depletion capacitors – voltage variable capacitors (varactors)
- Components of a PN junction:
 1. p-doped semiconductor – a semiconductor having atoms containing a lack of electrons (acceptors). The concentration of acceptors is N_A in atoms per cubic centimeter.
 2. n-doped semiconductor – a semiconductor having atoms containing an excess of electrons (donors). The concentration of these atoms is N_D in atoms per cubic centimeter.

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PN JUNCTIONS

Physical structure

- **PN junction** structure consists of
 - p-type semiconductor
 - n-type semiconductor
 - metal contact for connection
- As the PN junction implements the junction diode, its terminals are labeled anode and cathode.

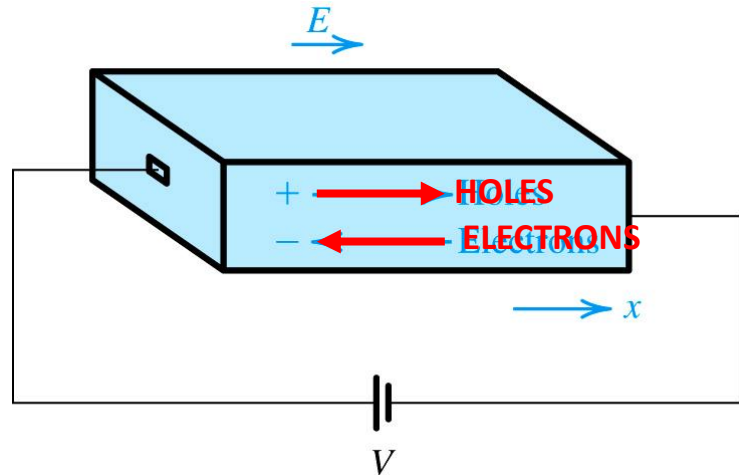


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PN JUNCTIONS

Physical structure

- An electric field E established in a bar of silicon causes the holes to drift in the direction of E and the free electrons to drift in the opposite direction. Both the hole and electron drift currents are in the direction of E .
- **bound charge**
 - charge of **opposite** polarity to free electrons / holes of a given material
 - **neutralizes the electrical charge** of these majority carriers
 - does not affect **concentration** gradients



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PN JUNCTIONS

Physical structure

- **p-type semiconductor**
 - Silicon is doped with element having a **valence of 3**.
 - To increase the concentration of **holes** (p).
 - One example is **boron**, which is an acceptor.
- **n-type semiconductor**
 - Silicon is doped with element having a **valence of 5**.
 - To increase the concentration of free **electrons** (n).
 - One example is **phosphorus**, which is a donor.

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PN JUNCTIONS

Physical structure

■ **p-type doped semiconductor**

- If N_A is much greater than n_i ...
 - concentration of acceptor atoms is N_A
- Then the **concentration of holes** in the p-type is defined as

$$\underbrace{(p_p)}_{\substack{\text{number} \\ \text{holes} \\ \text{in} \\ \text{p-type}}} \approx \underbrace{(N_A)}_{\substack{\text{number} \\ \text{acceptor} \\ \text{atoms}}}$$

■ **n-type doped semiconductor**

- If N_D is much greater than n_i ...
 - concentration of donor atoms is N_D
- Then the **concentration of electrons** in the n-type is defined as

$$\underbrace{(n_n)}_{\substack{\text{number} \\ \text{free} \\ \text{e-trons} \\ \text{in n-type}}} \approx \underbrace{(N_D)}_{\substack{\text{number} \\ \text{donor} \\ \text{atoms}}}$$

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PN JUNCTIONS

Physical structure

action: combine this with equation on previous slide

$$\underbrace{p_p}_{\substack{\text{number} \\ \text{of holes} \\ \text{in p-type}}} \times \underbrace{n_p}_{\substack{\text{number} \\ \text{of free} \\ \text{electrons} \\ \text{in p-type}}} = \underbrace{n_i^2}_{\substack{\text{number} \\ \text{of free} \\ \text{electrons} \\ \text{and holes} \\ \text{in thermal} \\ \text{equil.}}}$$

$$n_p \approx \frac{n_i^2}{N_A}$$

- Velocity of the holes:

μ_p =hole mobility
 E =electric field

$$v_{p\text{-drift}} = \mu_p E$$

action: combine this with equation on previous slide

$$\underbrace{p_n}_{\substack{\text{number} \\ \text{of holes} \\ \text{in n-type}}} \times \underbrace{n_n}_{\substack{\text{number} \\ \text{of free} \\ \text{electrons} \\ \text{in n-type}}} = \underbrace{n_i^2}_{\substack{\text{number} \\ \text{of free} \\ \text{electrons} \\ \text{and holes} \\ \text{in thermal} \\ \text{equil.}}}$$

$$p_n \approx \frac{n_i^2}{N_D}$$

μ_n =electron mobility
 E =electric field

$$v_{n\text{-drift}} = -\mu_n E$$

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PN JUNCTIONS

Drift current

- Substitute $v_p = \mu_p E$.
- Define current density as $J_p = I_p / A$.

I_p = current flow attributed to holes
 A = cross-sectional area of silicon
 q = magnitude of the electron charge
 p = concentration of holes
 $v_{p-drift}$ = drift velocity of holes

$$I_p = Aqp v_{p-drift}$$

I_p = current flow attributed to holes
 A = cross-sectional area of silicon
 q = magnitude of the electron charge
 p = concentration of holes
 μ_p = hole mobility
 E = electric field

$$I_p = Aqp \mu_p E$$

$$J_p = qp \mu_p E$$

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PN JUNCTIONS

Drift current

- conductivity** (σ) – relates total drift current density (J) and electrical field (E)
- resistivity** (ρ) – relates total drift current density (J) and electrical field (E)

Ohm's Law

$$J = \sigma E$$

$$\sigma = q(p\mu_p + n\mu_n)$$

$$J = E / \rho$$

$$\rho = \frac{1}{q(p\mu_p + n\mu_n)}$$

I_n = current flow attributed to electrons
 A = cross-sectional area of silicon
 q = magnitude of the electron charge
 n = concentration of free electrons
 μ_n = electron mobility
 E = electric field

$$I_n = -Aq v_{n-drift}$$

$$J_n = qn \mu_n E$$

$$J = J_p + J_n = \underbrace{q(p\mu_p + n\mu_n)}_{\text{this is conductivity } (\sigma)} E$$

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PN JUNCTIONS

J_p = current flow density attributed to holes
 q = magnitude of the electron charge
 D_p = diffusion constant of holes ($12\text{cm}^2/\text{s}$ for silicon)
 $p(x)$ = hole concentration at point x
 dp/dx = gradient of hole concentration

Drift current

hole diffusion current density : $J_p = -qD_p \frac{dp(x)}{dx}$

electron diffusion current density : $J_n = -qD_n \frac{dn(x)}{dx}$

J_n = current flow density attributed to free electrons
 D_n = diffusion constant of electrons ($35\text{cm}^2/\text{s}$ for silicon)
 $n(x)$ = free electron concentration at point x
 dn/dx = gradient of free electron concentration

A = cross-sectional area of silicon, q = magnitude of the electron charge,
 p = concentration of holes, n = concentration of free electrons,
 μ_p = hole mobility, μ_n = electron mobility, E = electric field

drift current density : $J_{\text{drift}} = J_{p\text{-drift}} + J_{n\text{-drift}} = q(p\mu_p + n\mu_n)E$

diffusion current density : $J_{\text{diff}} = J_{p\text{-diff}} + J_{n\text{-diff}} = -qD_p \frac{dp(x)}{dx} - qD_n \frac{dn(x)}{dx}$

D_p = diffusion constant of holes ($12\text{cm}^2/\text{s}$ for silicon), D_n = diffusion constant of electrons ($35\text{cm}^2/\text{s}$ for silicon),
 $p(x)$ = hole concentration at point x , $n(x)$ = free electron concentration at point x ,
 dp/dx = gradient of hole concentration, dn/dx = gradient of free electron concentration

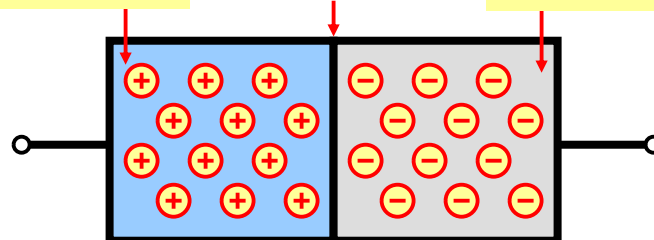
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PN JUNCTIONS

Operations

- The PN junction with no applied voltage

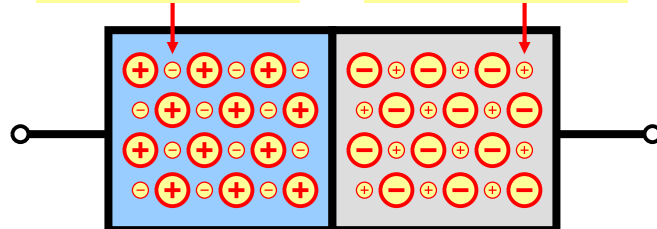
p -type semiconductor filled with holes junction n -type semiconductor filled with free electrons



negative bound charges

positive bound charges

- Bound charges are attracted (from environment) by free electrons and holes in the p -type and n -type semiconductors, respectively. They remain weakly "bound" to these majority carriers; however, they do not recombine.

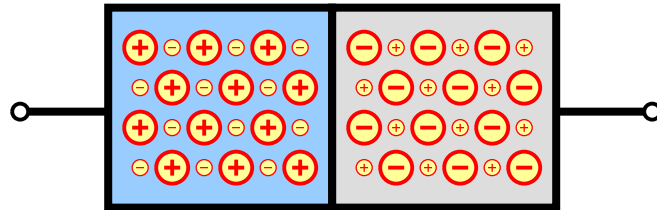


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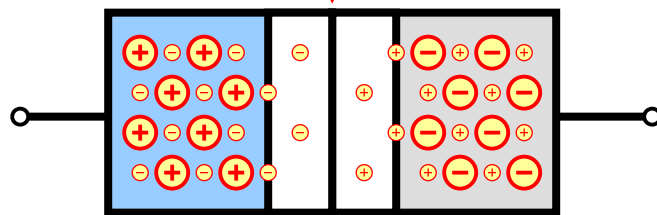
PN JUNCTIONS

Operations

- Diffusion begins. Those free electrons and holes which are closest to the junction will recombine and, essentially, eliminate one another
- The depletion region begins to form – as diffusion occurs and free electrons recombine with holes.



The depletion region is filled with "uncovered" bound charges – who have lost the majority carriers to which they were linked.

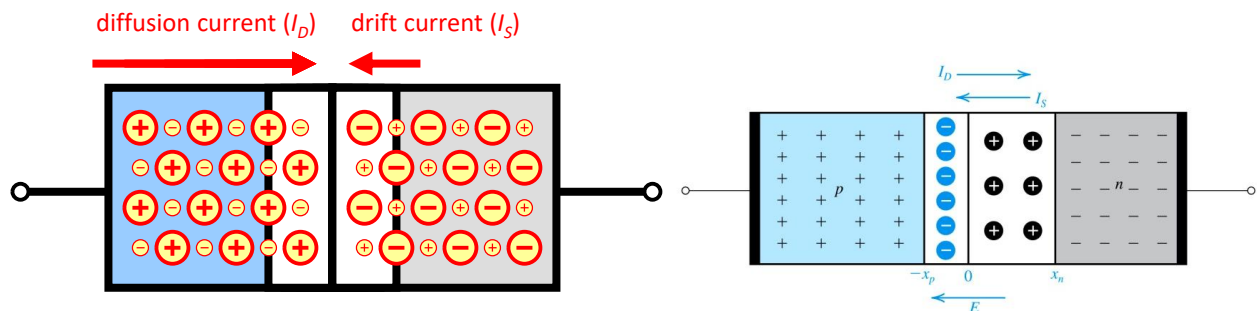


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PN JUNCTIONS

Operations

- The barrier voltage (V_0) is an electric field whose polarity opposes the direction of diffusion current (I_D). As the magnitude of V_0 increases, the magnitude of I_D decreases.



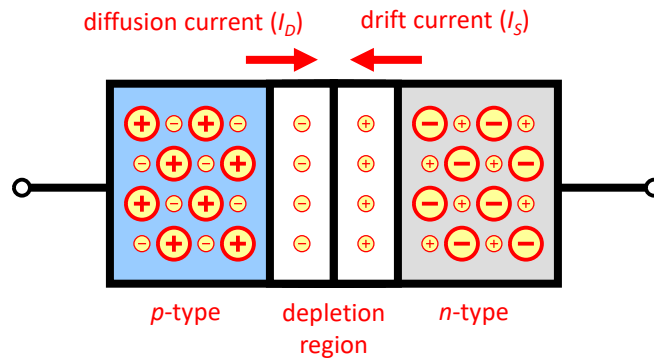
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PN JUNCTIONS

Operations

- Equilibrium is reached, and diffusion ceases, once the magnitudes of diffusion and drift currents equal one another – resulting in **no net flow**.

Once equilibrium is achieved, no net current flow exists ($I_{net} = I_D - I_S$) within the *pn*-junction while under open-circuit condition.



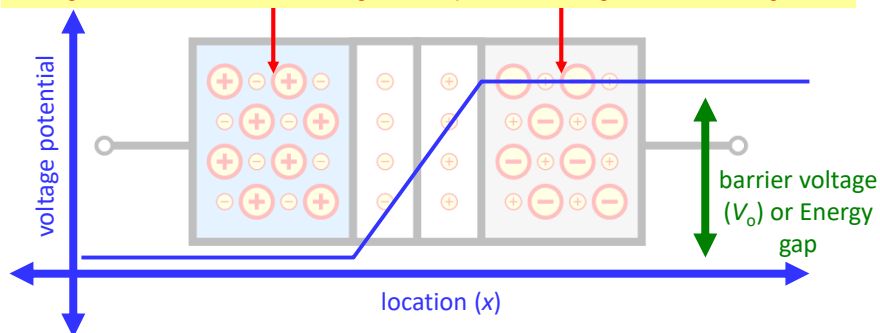
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PN JUNCTIONS

Operations

- The “uncovered” bound charges effect a voltage differential across the depletion region. The magnitude of this barrier voltage (V_0) differential grows, as diffusion continues.

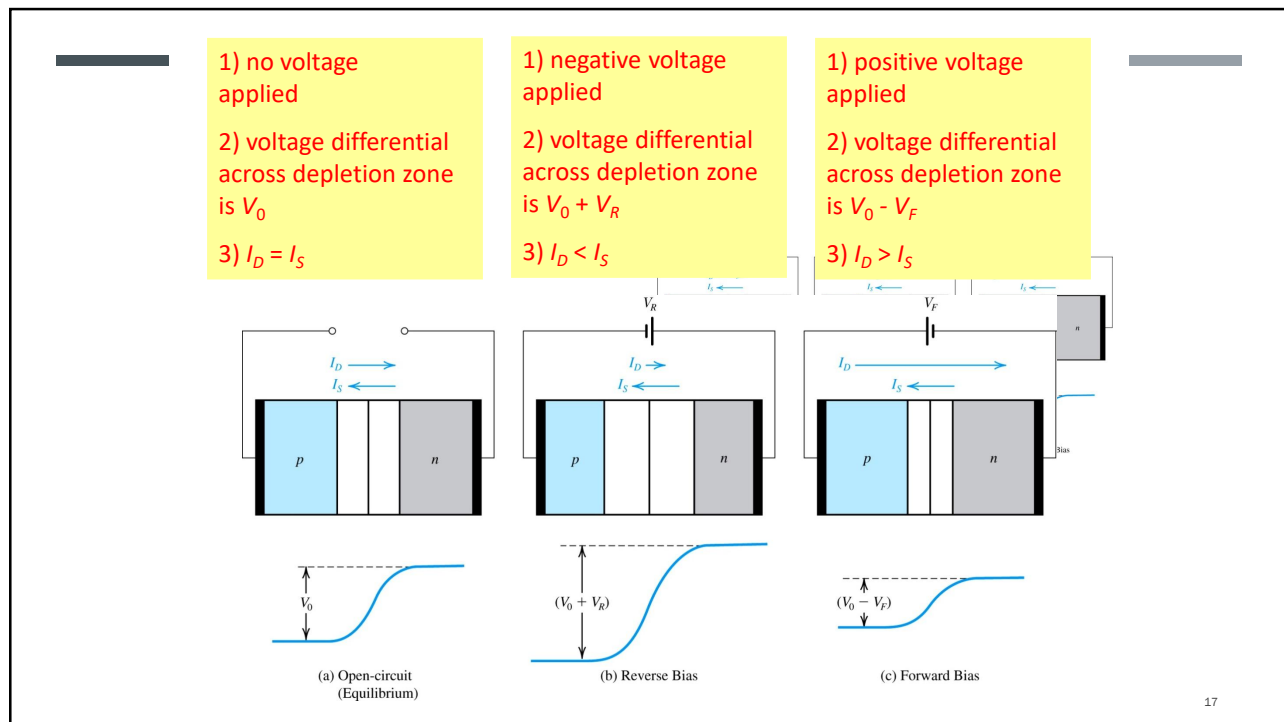
No voltage differential exists across regions of the *pn*-junction outside of the depletion region because of the neutralizing effect of positive and negative bound charges.



V_0 = barrier voltage
 V_T = thermal voltage
 N_A = acceptor doping concentration
 N_D = donor doping concentration
 n_i = concentration of free electrons...
 ...in intrinsic semiconductor

$$V_0 = V_T \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

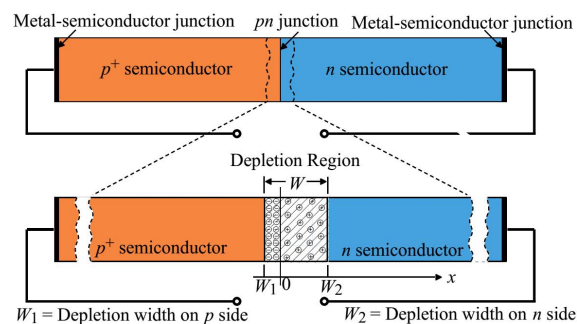
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PN JUNCTIONS

PN Junction



1. $W_1 = x_p$ = Depletion width on p side.
2. $W_2 = x_n$ = Depletion width on n side.
3. *Equilibrium conditions* are reached when:

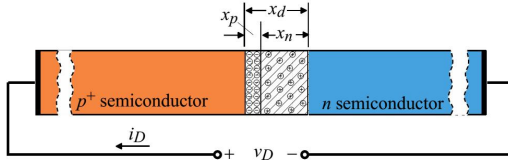
Current due to diffusion = Current due to drift

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PN JUNCTIONS

Graphical Characterization of the PN Junction

- Assume the PN junction is open-circuit.
- Width of depletion region: $x_d = x_p + x_n$

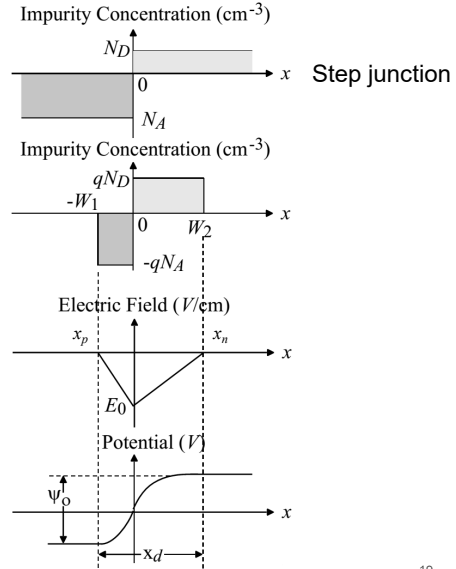
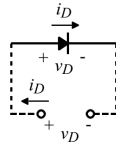


- Symbol for the PN junction (diode):

Built-in barrier potential, V_0 or ψ_0 :

$$\psi_0 = V_t \ln\left(\frac{N_A N_D}{n_i^2}\right) \quad V_t = \frac{kT}{q}$$

n_i is the intrinsic concentration of silicon.



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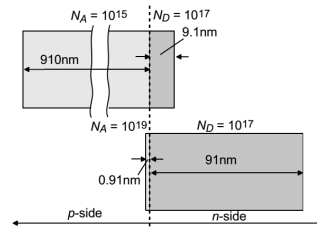
PN JUNCTIONS

Influence of Doping Level on the Depletion Regions

- Intuitively, one can see that the depletion regions are inversely proportional to the doping level. To achieve equilibrium, equal and opposite fixed charge on both sides of the junction are required. Therefore, the larger the doping, the smaller the depletion region on that side of the junction.
- P-type: holes $\rightarrow$ acceptor $\rightarrow$ acceptor concentration N_A
- N-Type: electrons $\rightarrow$ donor $\rightarrow$ donor concentration N_D
- The equations of that result are:

$$W_1 = \sqrt{\frac{2\epsilon(\psi_0 - v_D)}{qN_A(1 + \frac{N_A}{N_D})}} \propto \sqrt{\frac{1}{N_A}}$$

$$W_2 = \sqrt{\frac{2\epsilon(\psi_0 - v_D)}{qN_D(1 + \frac{N_D}{N_A})}} \propto \sqrt{\frac{1}{N_D}}$$



- Example: Assume that $v_D = 0$, $\psi_0 = 0.637V$ and $N_D = 10^{17}$ atoms/cm³. Find the p-side depletion region width if $N_A = 10^{15}$ atoms/cm³ and if $N_A = 10^{19}$ atoms/cm³:
 - For $N_A = 10^{15}$ atoms/cm³ the p-side depletion width is 0.90 μm .
 - For $N_A = 10^{19}$ atoms/cm³ the p-side depletion width is 0.9 nm.

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PN JUNCTIONS

Forward-Biased PN Junctions

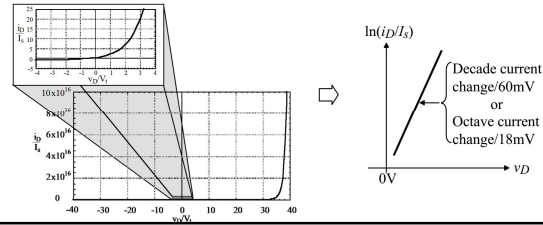
- When the PN junction is forward-biased, the potential barrier is reduced and significant current begins to flow across the junction. The diode voltage-current relationship that characterizes the PN junction is given by:

$$i_D = qA \left[\frac{D_p p_{n0}}{L_p} + \frac{D_n p_{p0}}{L_n} \right] \left[\exp\left(\frac{v_D}{V_t} - 1\right) \right] = I_s \left[\exp\left(\frac{v_D}{V_t} - 1\right) \right]$$

- Where i_s is saturation current:

$$i_s = qA \left[\frac{D_p p_{n0}}{L_p} + \frac{D_n p_{p0}}{L_n} \right] \approx \frac{qAD}{L} \frac{n_i^2}{N} = KT^3 \exp\left(\frac{-V_{G0}}{V_t}\right)$$

- Graphically, the i_D versus v_D characteristics are given as:



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PN JUNCTIONS

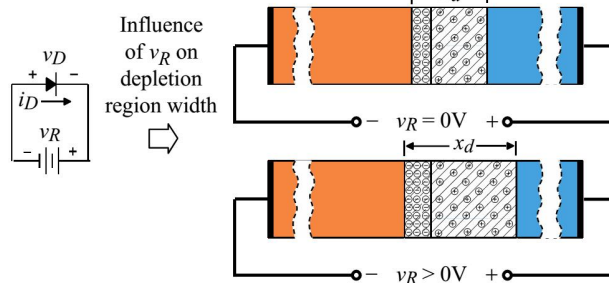
Reverse-Biased PN Junctions

- Depletion region:

$$x_d = x_p + x_n = W_1 + W_2$$

$$x_p = W_1 \propto \sqrt{v_R}$$

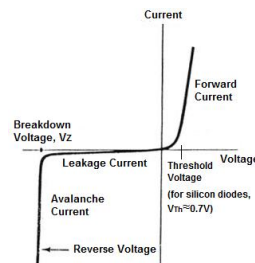
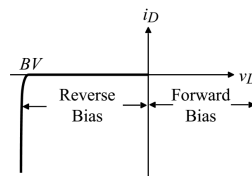
$$x_n = W_2 \propto \sqrt{v_R}$$



- Breakdown voltage (BV): is maximum reverse-bias voltage

In the reverse direction the current can be written as

$$i_D = \frac{-I_R}{1 - \left(\frac{v_R}{BV}\right)^n}$$



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PN JUNCTIONS

Breakdown Voltage as a Function of Doping

- It can be shown that: (BV = Breakdown Voltage)

$$BV \approx \frac{\epsilon_{si}(N_A + N_D)}{2qN_A N_D} E_{max}^2$$

where $E_{max} = 3 \times 10^5$ V/cm for silicon $\rightarrow$ BV = 58.2 V

- Example: Assume that $N_D = 10^{17}$ atoms/cm³. Find BV if $N_A = 10^{15}$ atoms/cm³ and if $N_A = 10^{19}$ atoms/cm³:

- $N_A = 10^{15}$ atoms/cm³: If $N_A \ll N_D$:

$$BV \approx \frac{\epsilon_{si}}{2qN_A} E_{max}^2 = \frac{1.04 \times 10^{-12} \cdot 9 \times 10^{10}}{2 \cdot 1.6 \times 10^{-19} \cdot 10^{15}} = 291 \text{ V}$$

- $N_A = 10^{19}$ atoms/cm³: If $N_A \gg N_D$:

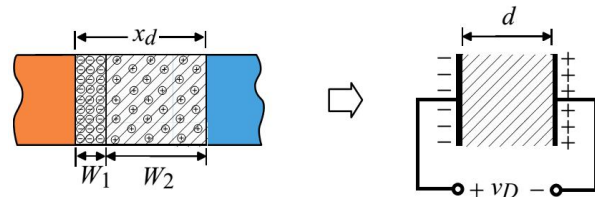
$$BV \approx \frac{\epsilon_{si}}{2qN_D} E_{max}^2 = \frac{1.04 \times 10^{-12} \cdot 9 \times 10^{10}}{2 \cdot 1.6 \times 10^{-19} \cdot 10^{17}} = 2.91 \text{ V}$$

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PN JUNCTIONS

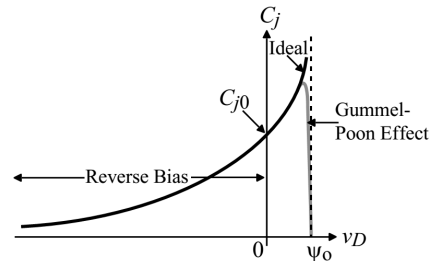
Depletion Capacitance

- Physical viewpoint of the depletion capacitance:



Depletion capacitance for a PN junction, when v_D is positive and approaches ψ_0 . Depletion capacitance approaches infinity.

$$\begin{aligned} C_j &= \frac{\epsilon_{si} A}{d} = \frac{\epsilon_{si} A}{W_1 + W_2} \\ &= \frac{\epsilon_{si} A}{\sqrt{\frac{2\epsilon_{si}(\psi_0 - v_D)}{q(N_D + N_A)}} \left[\sqrt{\frac{N_D}{N_A}} + \sqrt{\frac{N_A}{N_D}} \right]} \\ &= A \sqrt{\frac{\epsilon_{si} q N_A N_D}{2(N_A + N_D)}} \frac{1}{\sqrt{\psi_0 - v_D}} \\ &= \frac{C_{j0}}{\sqrt{1 - \frac{v_D}{\psi_0}}} \end{aligned}$$



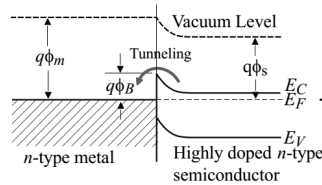
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PN JUNCTIONS

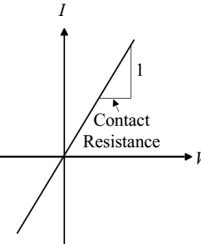
Metal-Semiconductor Junctions

- Ohmic Junctions: A metal-semiconductor junction formed by a highly doped semiconductor and metal.
- Schottky Junctions: A metal-semiconductor junction formed by a lightly doped semiconductor and metal.

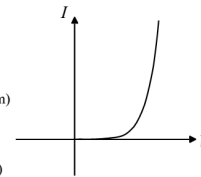
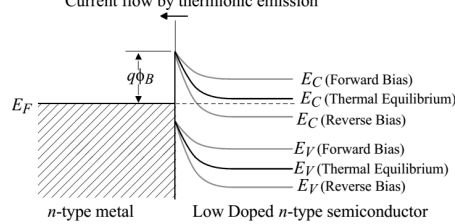
Energy band diagram



I-V Characteristics



Current flow by thermionic emission



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PN JUNCTIONS

Exercise 1:

- Find x_p , x_n , x_d , ψ_0 , C_{j0} , C_j for an applied voltage of -4V of a pn diode with a step junction, $N_A = 5 \times 10^{15}/\text{cm}^3$, $N_D = 10^{20}/\text{cm}^3$, and an area of $10\mu\text{m}$ by $10\mu\text{m}$.
- Solution:
- At room temperature, $k = 1.38 \times 10^{-23}$ J/K (Boltzmann's constant)

$$\psi_0 = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right) = 0.917V$$

$$x_n = W_1 = \left[\frac{2\epsilon_{Si}(\psi_0 - v_D)N_A}{qN_D(N_A + N_D)} \right]^{1/2} \cong 0$$

$$x_p = W_2 = - \left[\frac{2\epsilon_{Si}(\psi_0 - v_D)N_D}{qN_A(N_A + N_D)} \right]^{1/2} = 1.128\mu\text{m}$$

- $x_d = x_n + x_p = 1.128\mu\text{m}$
- $C_{j0} = 20.3\text{fF}$ at voltage of -4V and $C_j = 9.18\text{fF}$

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PN JUNCTIONS

Exercise 2:

- $N_A = 5 \times 10^{15}/\text{cm}^3$, $N_D = 10^{20}/\text{cm}^3$, $D_N = 20 \text{ cm}^2/\text{s}$, $D_P = 10 \text{ cm}^2/\text{s}$, $L_N = 10 \mu\text{m}$, $L_P = 5 \mu\text{m}$, $A = 10 \mu\text{m}^2$.
Calculate the saturation current of pn junction.

Solution:

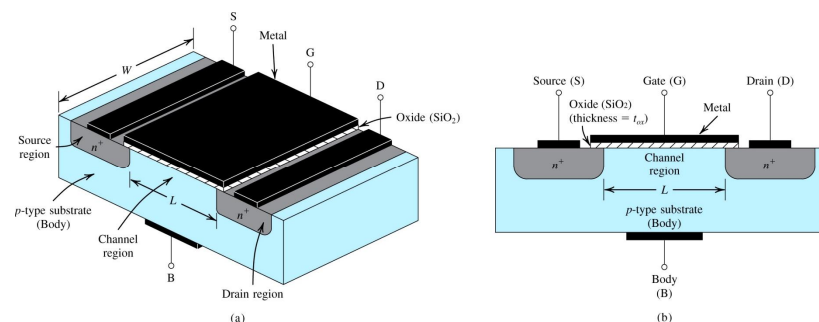
$$I_S = qA \left[\frac{D_p p_{n0}}{L_p} + \frac{D_n p_{p0}}{L_n} \right]$$

- p_{n0} is calculated from $n_i^2/N_D = 2.103/\text{cm}^3$;
- n_{p0} is calculated from $n_i^2/N_A = 4.205 \times 10^4/\text{cm}^3$;
- The saturation current $I_s = 1.346 \times 10^{-15} \text{ A} = 1.346 \text{ fA}$

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MOS TRANSISTOR

Device Structure



Physical structure of the enhancement-type NMOS transistor

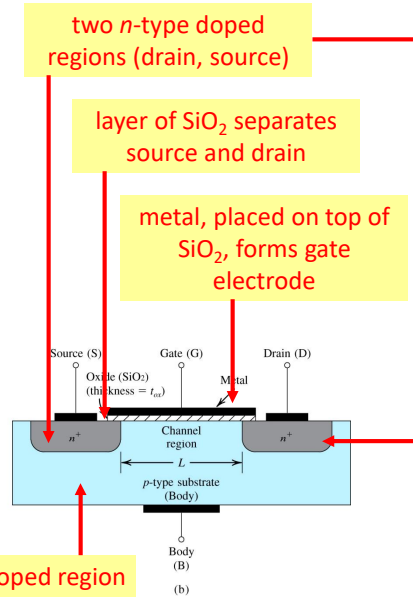
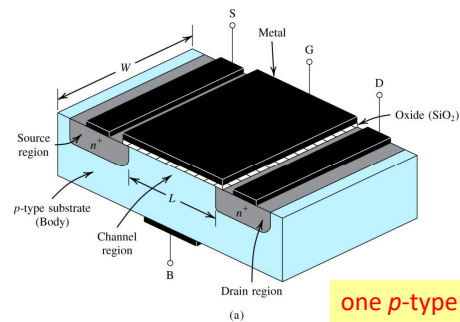
(a) perspective view, (b) cross-section.

Note that typically $L = 0.03 \mu\text{m}$ to $1 \mu\text{m}$, $W = 0.1 \mu\text{m}$ to $100 \mu\text{m}$, and the thickness of the oxide layer (t_{ox}) is in the range of 1 to 10 nm .

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MOS TRANSISTOR

Device Structure



one p-type doped region

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MOS TRANSISTOR

Device Structure

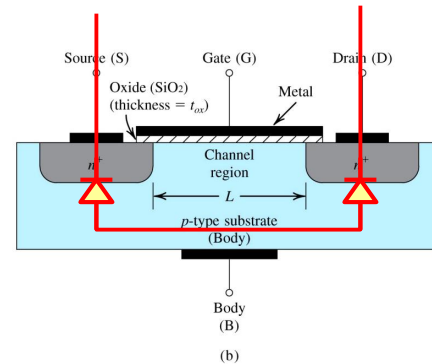
- The **name** MOSFET is derived from its physical structure.
- However, many MOSFET's do not actually use any "metal", **polysilicon is used instead**.
 - "This" has no effect on modeling / operation as described here.
- Another name for MOSFET is Insulated Gate FET, or **IGFET**.
- The **device is composed of two pn-junction**, however they maintain reverse biasing at all times.
 - Drain will always be at positive voltage with respect to source.
- We will not consider conduction of current in this manner.

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MOS TRANSISTOR

Device Structure

- With zero voltage applied to gate, **two back-to-back diodes** exist in series between drain and source.
- "They" **prevent current conduction (no conduction)** from drain to source when a voltage v_{DS} is applied.
 - yielding very high resistance (10^{12} **ohms**)

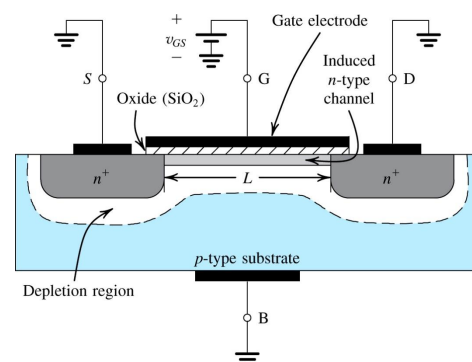


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MOS TRANSISTOR

Device Structure

- What happens if (1) **source and drain are grounded** and (2) **positive voltage is applied to gate**? Refer to figure to right.
 - step #1:** v_{GS} is applied to the gate terminal, causing a **positive build up of positive charge** along metal electrode.
 - step #2:** This "build up" causes **free holes to be repelled** from region of p -type substrate under gate.



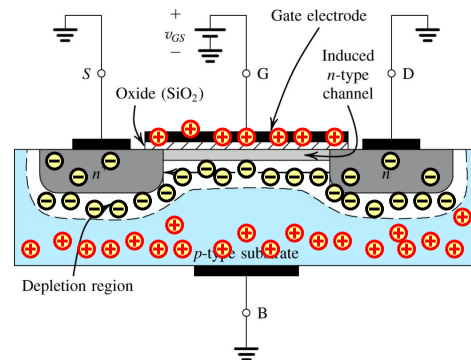
The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate

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MOS TRANSISTOR

Device Structure

- **Q:** What happens if (1) source and drain are grounded and (2) positive voltage is applied to gate? Refer to figure to right.
- **step #3:** This “migration” results in the uncovering of negative **bound charges**, originally neutralized by the free holes
- **step #4:** The positive gate voltage also **attracts electrons** from the n^+ source and drain regions into the channel.



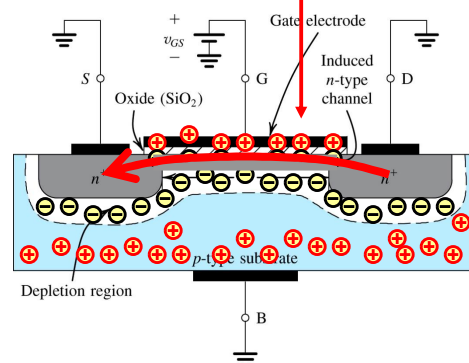
The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate

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MOS TRANSISTOR

Device Structure

- **Q:** What happens if (1) source and drain are grounded and (2) positive voltage is applied to gate? Refer to figure to right.
- **step #5:** Once a sufficient number of “these” electrons accumulate, an n -region is created...
 - ...connecting the source and drain regions
- **step #6:** This **provides path for current (conduction)** flow between D and S .
- **Inversion layer:** n -channel is formed under the gate oxide. It is inverted from p -type to n -type due to electron accumulation.

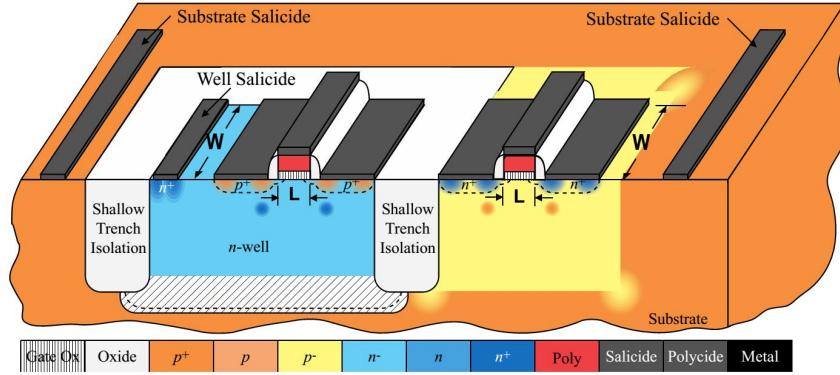


The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate

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MOS TRANSISTOR

Physical Structure of MOS Transistors in an n-well Technology



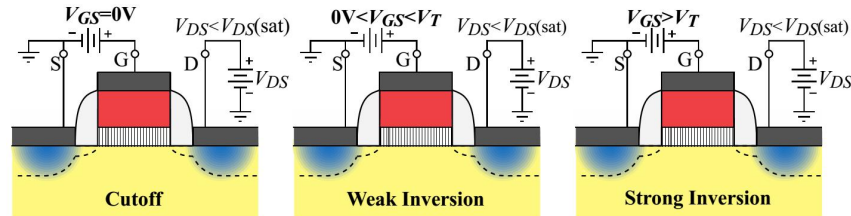
- Width (W) of the MOSFET = Width of the source/drain diffusion
- Length (L) of the MOSFET = Width of the polysilicon gate between the S/D diffusions
- Note that the MOSFET is isolated from the well/substrate by reverse biasing the resulting pn junction

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MOS TRANSISTOR

Enhancement MOSFETs

- The channel of an enhancement MOSFET is formed when the proper potential is applied to the gate of the MOSFET. This potential inverts the material immediately below the gate to the same type of impurity as the source and drain forming the channel.



- V_T = Gate-bulk work function (ϕ_{MS}) + voltage to change the surface potential ($-2\phi_F$) + voltage to offset the channel-bulk depletion charge ($-Q_b/C_{ox}$) + voltage to compensate the undesired interface charge ($-Q_{ss}/C_{ox}$)

$$V_T = \phi_{MS} - 2\phi_F - \frac{Q_{b0}}{C_{ox}} - \frac{Q_{ss}}{C_{ox}} - \frac{Q_b - Q_{b0}}{C_{ox}} = V_{T0} + \gamma(\sqrt{|-2\phi_F + v_{SB}|} - \sqrt{|-2\phi_F|})$$

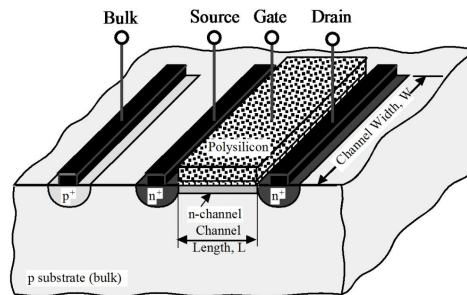
$$\text{where } V_{T0} = \phi_{MS} - 2\phi_F - \frac{Q_{b0}}{C_{ox}} - \frac{Q_{ss}}{C_{ox}}, \quad \gamma = \frac{\sqrt{2q\epsilon_{si}N_A}}{C_{ox}} \quad \text{and} \quad Q_b \approx \sqrt{2qN_A\epsilon_{si}(|-2\phi_F + v_{SB}|)}$$

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MOS TRANSISTOR

Depletion Mode MOSFETs

- The channel is diffused into the substrate so that a channel exists between the source and drain with no external gate potential.



- The threshold voltage for a depletion mode NMOS transistor will be negative (a negative gate potential is necessary to attract enough holes underneath the gate to cause this region to invert to p-type material).

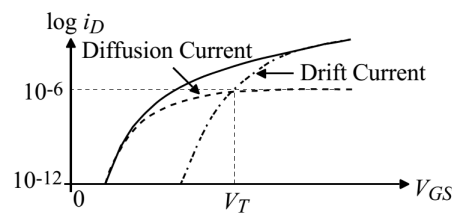
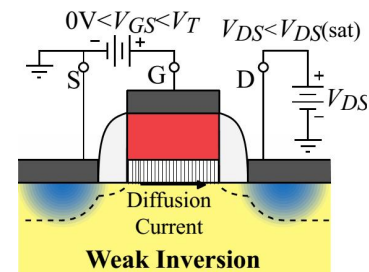
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MOS TRANSISTOR

Weak Inversion Operation

- Weak inversion operation occurs when the applied gate voltage is below V_T and occurs when the surface of the substrate beneath the gate is weakly inverted.
- Regions of operation according to the surface potential, ϕ_s .
 - $\phi_s < \phi_F$: Substrate not inverted
 - $\phi_F < \phi_s < 2\phi_F$: Channel is weakly inverted (diffusion current)
 - $2\phi_F < \phi_s$: Strong inversion (drift current)

- Drift current vs. diffusion current in a MOSFET:

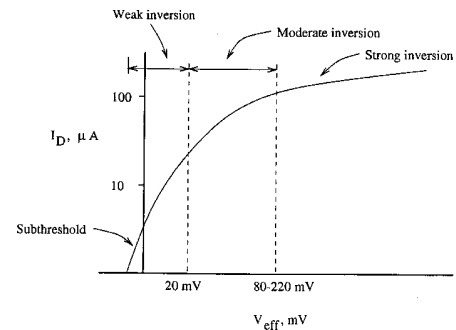


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MOS TRANSISTOR

Strong Inversion Operation

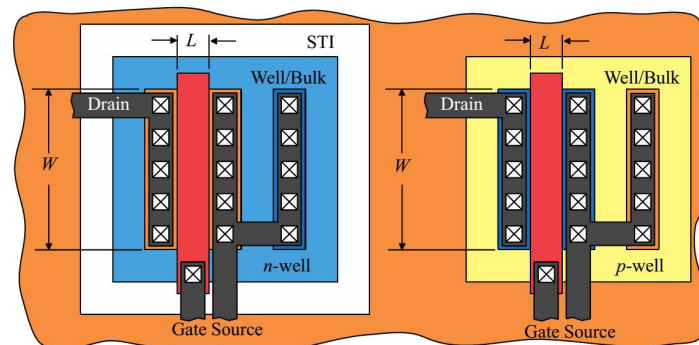
- At strong inversion, the electron concentration at the surface becomes equal to N_A . This happens when $\phi_s \geq 2\phi_F$ and the applied potential for this 0.7 V .
- The externally applied potential is equal to the voltage drop across the oxide layer plus the voltage drop across the semiconductor.



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MOS TRANSISTOR

Layout of a Single MOS Transistor

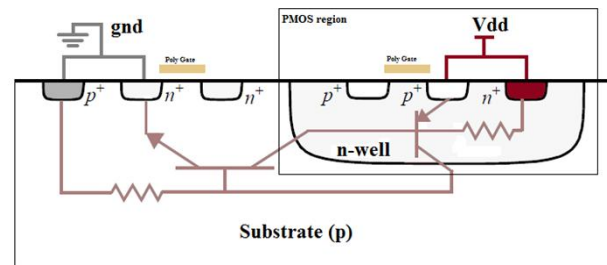


- Comments:
 - Make sure to contact the source and drain with multiple contacts to evenly distribute the current flow under the gate.
 - Minimize the area of the source and drain to reduce bulk-source/drain capacitance.

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LATCH-UP ISSUE

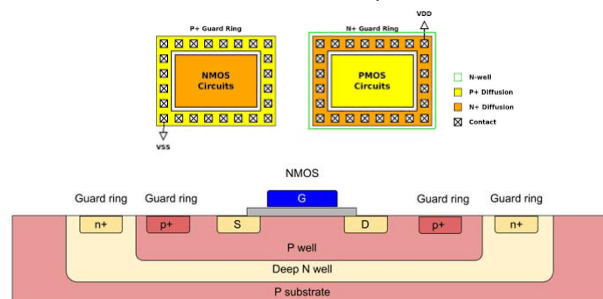
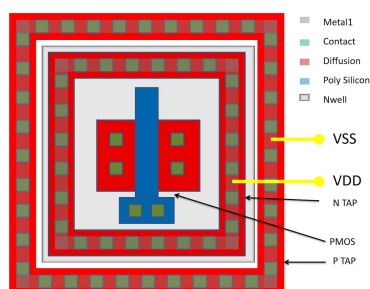
- A latch-up is a type of short circuit which can occur in an integrated circuit (IC).
- It is the phenomena of activating the parasitic BJTs in a CMOS circuit which forms a low impedance path between the power and ground terminals.
- This low impedance path draws a large current and heats up the IC (Integrated Chip) which cause permanent damage of IC.



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LATCH-UP ISSUE

- To prevent latch-up, we can use the following techniques:
 - Surrounding PMOS and NMOS transistors with an insulating oxide layer (trench) or using Guard Ring when doing layout.. This breaks parasitic SCR structure.
 - Latchup Protection Technology circuitry which shuts off the device when latchup is detected.

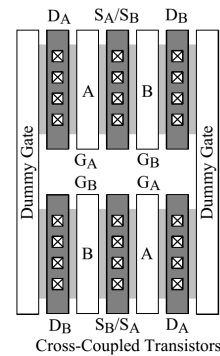
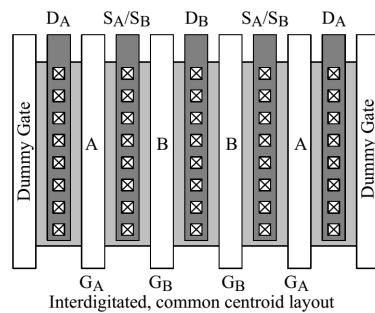


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MOS TRANSISTOR

Thermal and Stress Effects

- Oxide gradients – use common centroid geometry layout
- Stress gradients – use proper location and common centroid geometry layout
- Thermal gradients – keep transistors well away from power devices and use common centroid geometry layout with interdigitated transistors
- Examples of Common Centroid Interdigitated transistor layout:



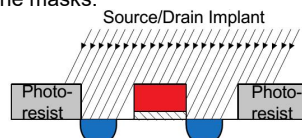
43

MOS TRANSISTOR

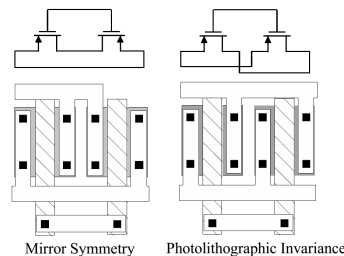
MOS Transistor Layout

- Photolithographic Invariance (PLI) are transistors that exhibit identical orientation. PLI comes from optical interactions between the UV light and the masks.

- Simple illustration of PLI:



- Examples of the layout of matched MOS transistors:
1) Examples of mirror symmetry and photolithographic invariance.



Notes:

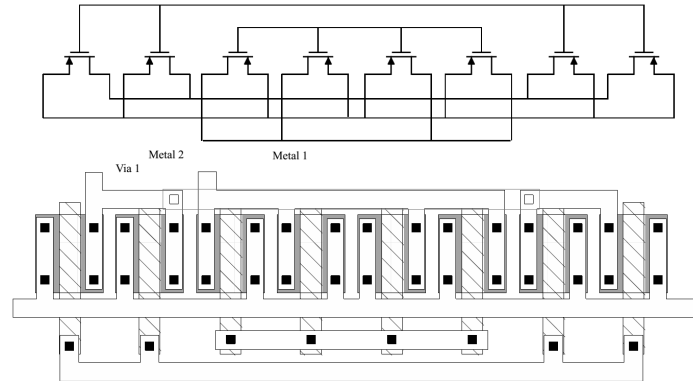
- The ratio between 2 similar components on the same integrated circuit can be controlled to better than 1% or 0.1%.
- Devices specifically constructed to obtain a known, constant ratio are called **matched devices**.

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MOS TRANSISTOR

MOS Transistor Layout

- Examples of the layout of matched MOS transistors:
 - Two transistors sharing a common source and laid out to achieve both photolithographic invariance and common centroid

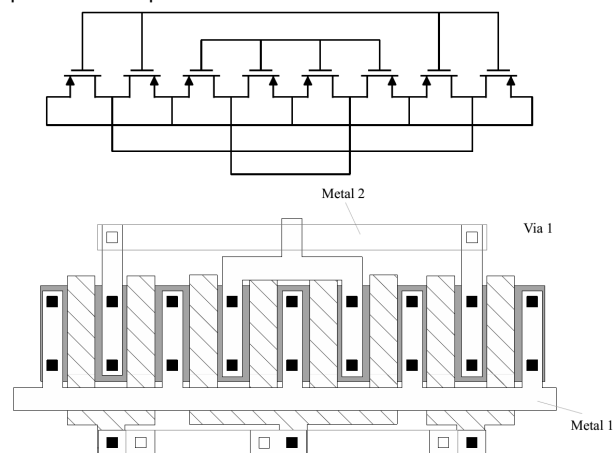


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MOS TRANSISTOR

MOS Transistor Layout

- Examples of the layout of matched MOS transistors:
 - Compact layout of the previous example

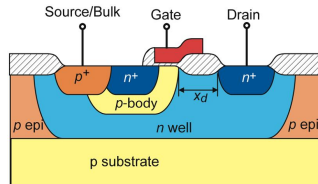


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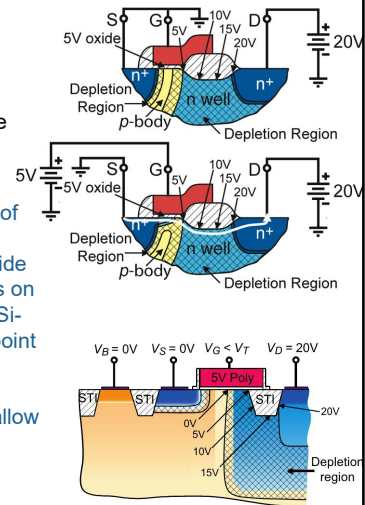
HIGH VOLTAGE CMOS TRANSISTORS

High Voltage Architectures

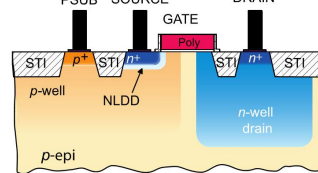
- To get extended voltage transistors, make *depletion region* larger
- The objective is to create a lightly doped, extended drain region where the high voltage of the drain can drop down to a level that will not cause the gate oxide to breakdown.
- LOCOS Architecture:



LOCOS (LOCAl Oxidation of Silicon): a microfabrication process where silicon dioxide is formed in selected areas on a silicon wafer having the Si-SiO₂ interface at a lower point than the rest of the silicon surface. As of 2008 it was largely superseded by Shallow Trench Isolation (STI).



- DSM Architecture:
DSM (Deep Sub Micron)



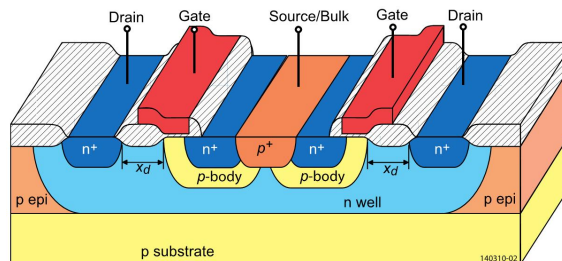
p ⁺ (≥10 ¹⁹)	p-well (5×10 ¹⁷)	HV p-well (10 ¹⁷)	p ⁺ (10 ¹⁶)	p-epi (10 ¹⁵)	n ⁻ (5×10 ¹⁴)	n ⁻ (10 ¹⁵)	HV n-well (10 ¹⁷)	n-well (5×10 ¹⁷)	n ⁺ (≥10 ²⁰)
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HIGH VOLTAGE CMOS TRANSISTORS

Lateral DMOS (LDMOS) Using LOCOS CMOS Technology

- The LDMOS (Laterally-Diffused MOS) structure is designed to provide sufficient lateral dimension and to prevent oxide breakdown by the higher drain voltages. It is used in power amplifiers.
- These transistors are often fabricated on p/p⁺ silicon epitaxial layers. One possible implementation using LOCOS technology:



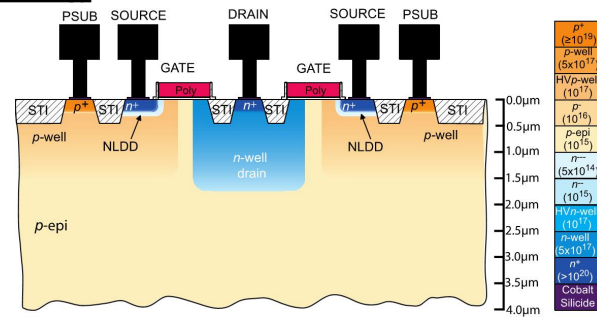
- Structure is symmetrical about the source/bulk contact
- Channel is formed in the p region under the gates
- The lightly doped n region between the drain side of the channel and the n⁺ drain contact (x_d) increases the depletion region width on the drain side of the channel/drain pn junction resulting in larger values of V_{DS} .
- Drain voltage can be 20-100V depending on the spacing and doping.

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HIGH VOLTAGE CMOS TRANSISTORS

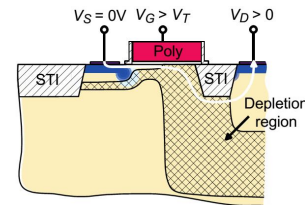
Lateral DMOS (LDMOS) Using LOCOS CMOS Technology

- Cross-section of an N-LDMOS using DSM technology:



- Differences between an N-LDMOS and NMOS:

- Asymmetry
- Non-uniform channel
- Current flow (not all at the surface)
- No self-alignment (larger drain-gate overlap capacitance)
- Note the extended drift region on the drain side of the channel

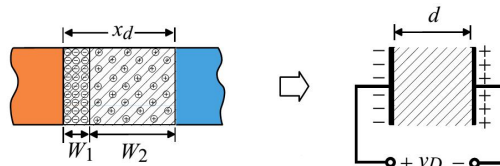


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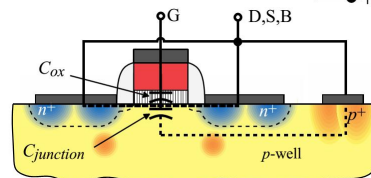
CAPACITORS

Types of Capacitors for CMOS Technology

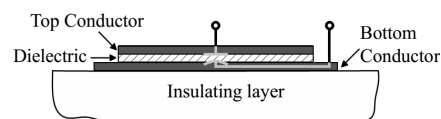
- PN junction (depletion) capacitors



- MOSFET gate capacitors



- Conductor-insulator-conductor capacitors



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CAPACITORS

Characterization of Capacitors

1. Losses in a capacitor characterized by the quality factor Q of a capacitor is a measure of the imaginary to real part of the impedance or admittance

$$Q = \frac{1}{\omega C R_s} = \omega C R_p$$

where R_p is the equivalent resistance in parallel with the capacitor, C , and R_s is the electrical series resistance (ESR) of the capacitor, C .

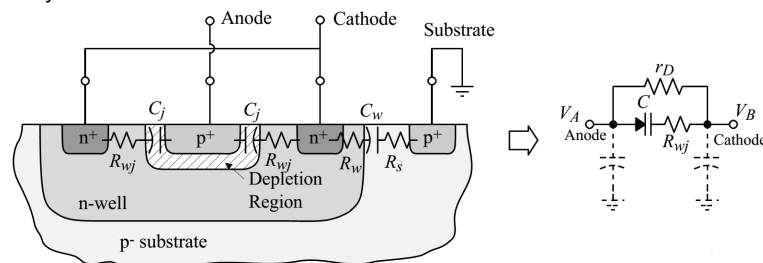
2. Parasitic capacitors to ground from each node of the capacitor.
3. The density of the capacitor in Farads/area.
4. The absolute and relative accuracies of the capacitor.
5. The C_{max}/C_{min} ratio which is the largest value of capacitance to the smallest when the capacitor is used as a variable capacitor (varactor).
6. The variation of a variable capacitance with the control voltage.
7. Linearity, $q = Cv$.

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PN JUNCTION CAPACITORS

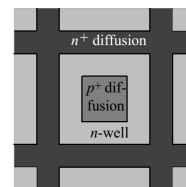
PN Junction Capacitors in a Well

- Generally made by diffusion into the well.



Layout:

- Minimize the distance between the p+ and n+ diffusions.
- Two different versions have been tested.
 1. Large islands – 9μm on a side
 2. Small islands – 1.2μm on a side



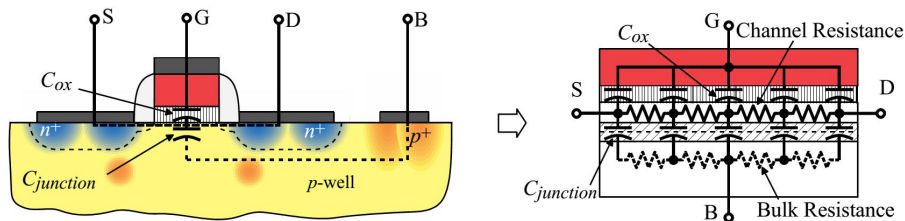
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MOSFET GATE CAPACITORS

MOSFET Gate Capacitor Structure

- The MOSFET gate capacitors have the gate as one terminal of the capacitor and some combination of the source, drain, and bulk as the other terminal.
- In the model of the MOSFET gate capacitor shown below, the gate capacitance is really two capacitors in series depending on the condition of the channel.

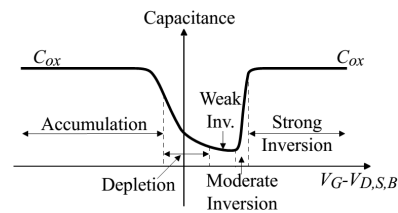
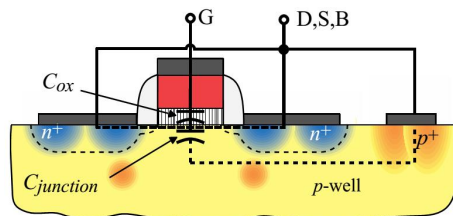
$$C_{\text{gate}} = \frac{1}{\frac{1}{C_{\text{ox}}} + \frac{1}{C_j}}$$



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MOSFET GATE CAPACITORS

MOSFET Gate Capacitor as a function of V_{GS} with $D=S=B$



- Operation: In this configuration, the MOSFET gate capacitor has 5 regions of operation as V_{GS} is varied. They are:

- Accumulation
- Depletion
- Weak inversion
- Moderate inversion
- Strong inversion

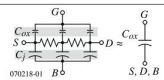
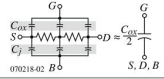
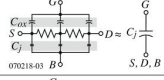
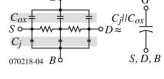
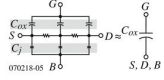
For the first four regions, the gate capacitance is the series combination of C_{ox} and C_j given as

$$C_{\text{gate}} = \frac{1}{\frac{1}{C_{\text{ox}}} + \frac{1}{C_j}}$$

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MOSFET GATE CAPACITORS

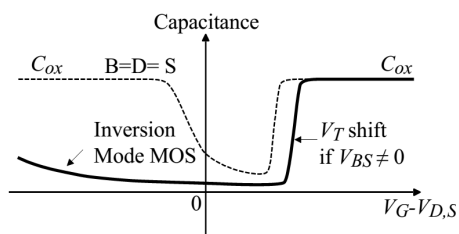
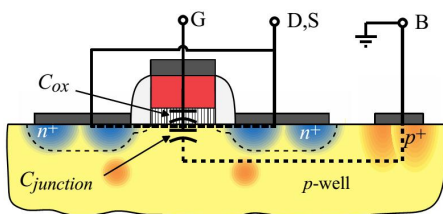
Use of a 3 Segment Model to Explain the Gate Capacitor Variation

Region	Channel R	C_{ox} and C_j	C_{gate}	3-Segment Model
Accumulation	Large	In series and $C_j > C_{ox}$	$C_{gate} \approx C_{ox}$	
Depletion	Large	In series and $C_j \approx C_{ox}$	$C_{gate} \approx 0.5C_{ox} \approx 0.5C_j$	
Weak Inversion	Large	In series and $C_j < C_{ox}$	$C_{gate} \approx C_j$	
Moderate Inversion	Moderate	In series and $C_j < C_{ox}$	$C_j < C_{gate} < C_{ox}$	
Strong Inversion	Small	In parallel and $C_j < C_{ox}$	$C_{gate} \approx C_{ox}$	

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MOSFET GATE CAPACITORS

MOSFET Gate Capacitor as a function of V_{GS} with Bulk Fixed (Inversion Mode)



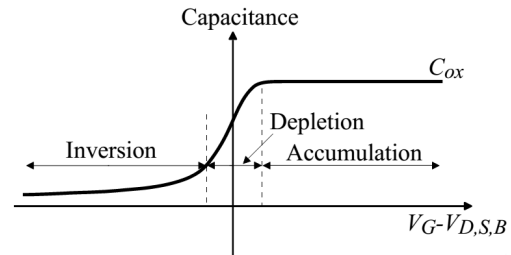
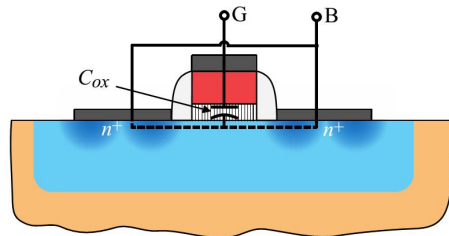
Conditions:

- $D = S, B = V_{SS}$
- Accumulation region removed by connecting bulk to ground
- Nonlinear
- Channel resistance: $R_{on} = \frac{L}{12K_P'(V_{BG} - |V_T|)}$
- LDD transistors will give lower Q because of the increased series resistance

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MOSFET GATE CAPACITORS

Accumulation Mode NMOS Gate Capacitor



Conditions:

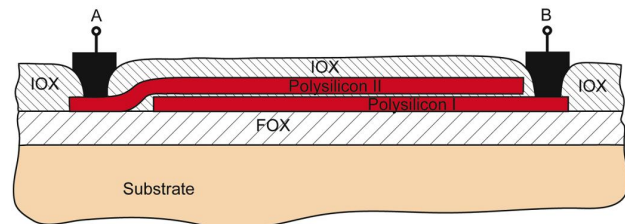
- Build the NMOS in a n-well or the PMOS in a p-well – channel is present with no bias
- Implements a variable capacitor with a larger transition region between the maximum and minimum values.
- Reasonably linear capacitor for values of $V_G - V_{D,S,B} > 0$

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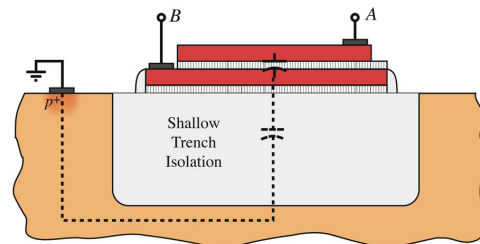
CONDUCTOR-INSULATOR-CONDUCTOR CAPACITORS

Polysilicon-Oxide-Polysilicon (Poly-Poly) Capacitors

- LOCOS Technology: A very linear capacitor with minimum bottom plate parasitic



- DSM Technology: A very linear capacitor with small bottom plate parasitic

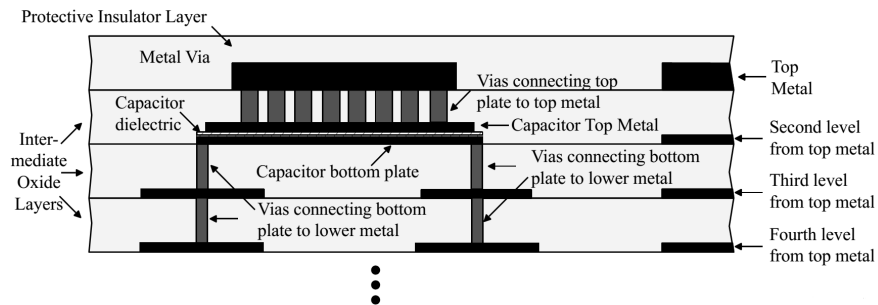


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CONDUCTOR-INSULATOR-CONDUCTOR CAPACITORS

Metal-Insulator-Metal (MiM) Capacitors (MIMCAP)

- In some processes, there is a thin dielectric between a metal layer and a special metal layer called “capacitor top metal”. Typically, the capacitance is around $1\text{fF}/\mu\text{m}^2$ and is at the level below top metal.
- Good matching is possible with low parasitics.

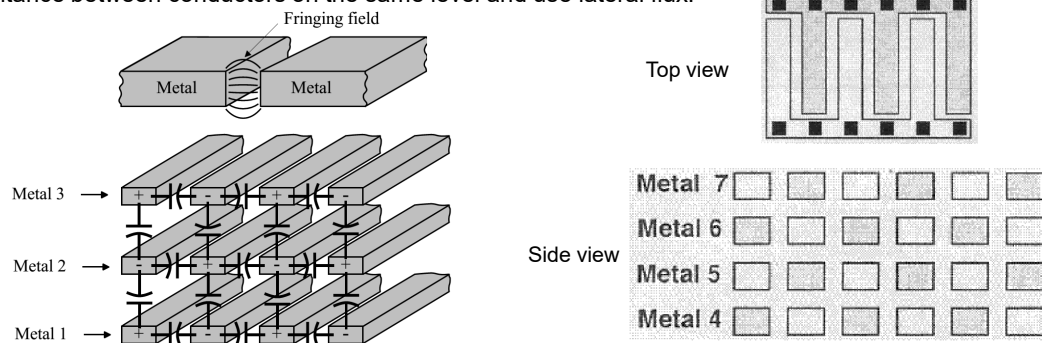


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CONDUCTOR-INSULATOR-CONDUCTOR CAPACITORS

Metal-Insulator-Metal Capacitors – Lateral and Vertical Flux

- Capacitance between conductors on the same level and use lateral flux.



- These capacitors are sometimes called fractal capacitors because the fractal patterns are structures that enclose a finite area with a near-infinite perimeter.
- The capacitor/area can be increased by a factor of 10 over vertical flux capacitors.

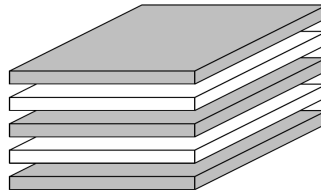
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CONDUCTOR-INSULATOR-CONDUCTOR CAPACITORS

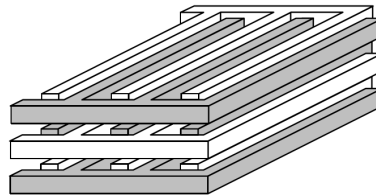
More Detail on Horizontal Metal Capacitors[†]

- Some of the possible metal capacitor structures include:

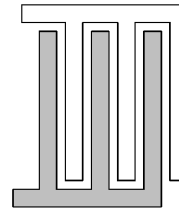
- Horizontal parallel plate (HPP)



- Parallel wires (PW)



Lateral View



Top View

[†] R. Aparicio and A. Hajimiri, "Capacity Limits and Matching Properties of Integrated Capacitors, IEEE J. of Solid-State Circuits, vol. 37, no. 3, March 2002, pp. 384-393.

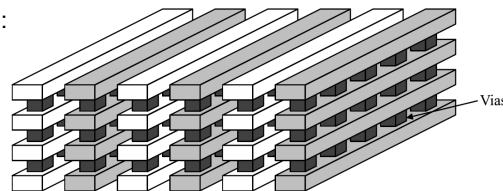
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CONDUCTOR-INSULATOR-CONDUCTOR CAPACITORS

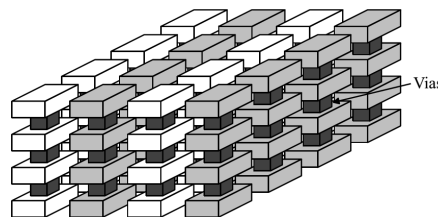
Horizontal Metal Capacitors - Continued

- Some of the possible metal capacitor structures include:

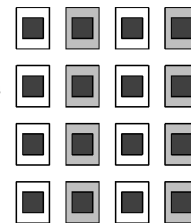
- Vertical parallel plates (VPP):



- Vertical bars (VB):



Lateral View



Top View

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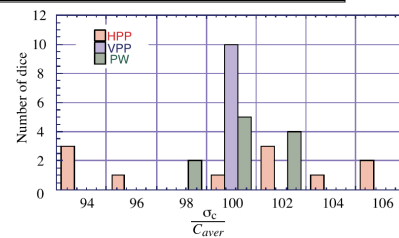
CONDUCTOR-INSULATOR-CONDUCTOR CAPACITORS

Horizontal Metal Capacitors - Continued

- Experimental results for a digital CMOS process with 7 layers of metal, $L_{min} = 0.24\mu\text{m}$, $t_{ox} = 0.7\mu\text{m}$ and $t_{metal} = 0.53\mu\text{m}$ for the bottom 5 layers of metal. All capacitors = 1pF.

Structure (1 pF)	Cap. Density (aF/ μm^2)	$C_{aver.}$ (pF)	Area (μm^2)	Cap. Enhanc ement	Std. Dev. (fF)	$\frac{\sigma}{C_{aver.}}$	$f_{res.}$ (GHz)	$Q @$ 1 GHz	Break- down (V)
VPP	1512.2	1.01	670	7.4	5.06	0.0050	> 40	83.2	128
VB	1281.3	1.07	839.7	6.3	14.19	0.0132	37.1	48.7	124
HPP	203.6	1.09	5378	1.0	26.11	0.0239	21	63.8	500
MIM	1100	1.05	960.9	5.4	-	-	11	95	-

- Histogram of the capacitance distribution:
- Result: The horizontal metal capacitors have a matching accuracy that is equivalent of the better capacitors: poly-poly and MIM.



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DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Capacitor Errors

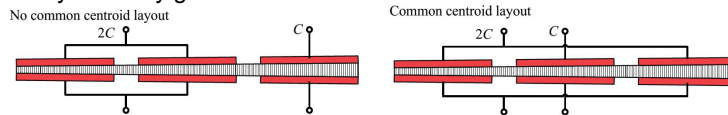
1. Dielectric gradients
2. Edge effects
3. Process biases
4. Parasitics
5. Voltage dependence
6. Temperature dependence

64

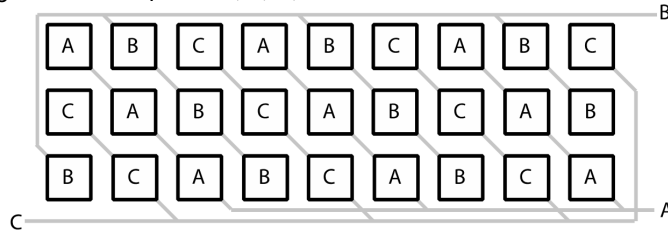
DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Capacitor Errors - Oxide Gradients

- Error due to a variation in dielectric thickness across the wafer.
- Common centroid layout - only good for one-dimensional errors:



- An alternate approach is to layout numerous repetitions and connect them randomly to achieve a statistical error balanced over the entire area of interest.
- Improved matching of three components, A, B, and C:

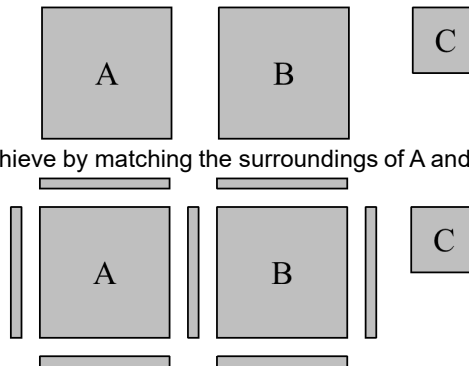


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DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Capacitor Errors - Edge Effects

- There will always be a randomness on the definition of the edge.
- However, etching can be influenced by the presence of adjacent structures.
- For example: Matching of A and B are disturbed by the presence of C



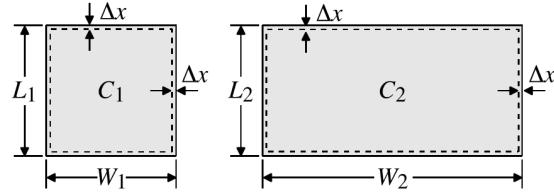
- Improved matching achieved by matching the surroundings of A and B.

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DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Process Bias on Capacitors

- Consider the following two capacitors:
- If $L_1 = L_2 = 2\mu\text{m}$, $W_2 = 2W_1 = 4\mu\text{m}$ and $\Delta x = 0.1\mu\text{m}$, the ratio of C_2 to C_1 can be written as,



$$\frac{C_2}{C_1} = \frac{(L_2 - 2\Delta x)(W_2 - 2\Delta x)}{(L_1 - 2\Delta x)(W_1 - 2\Delta x)} = \frac{(2 - 0.2)(4 - 0.2)}{(2 - 0.2)(2 - 0.2)} = \frac{3.8}{1.8} = 2.11 \rightarrow 5.6\% \text{ error in matching}$$

- How can this matching error be reduced? The capacitor ratios in general can be expressed as

$$\frac{C_2}{C_1} = \frac{(L_2 - 2\Delta x)(W_2 - 2\Delta x)}{(L_1 - 2\Delta x)(W_1 - 2\Delta x)} = \frac{W_2}{W_1} \left(\frac{1 - \frac{2\Delta x}{W_2}}{1 - \frac{2\Delta x}{W_1}} \right) \approx \frac{W_2}{W_1} \left(1 - \frac{2\Delta x}{W_2} \right) \left(1 + \frac{2\Delta x}{W_1} \right) \approx \frac{W_2}{W_1} \left(1 - \frac{2\Delta x}{W_2} + \frac{2\Delta x}{W_1} \right)$$

- Therefore, if $W_2 = W_1$, the matching error should be minimized. The best matching results between two components are achieved when their geometries are identical.

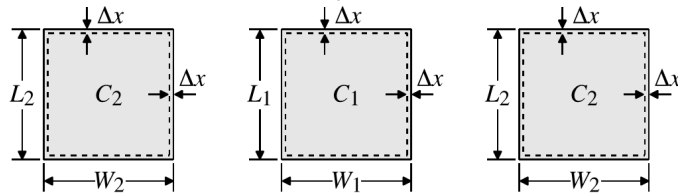
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DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Replication Principle

- Based on the previous result, a way to minimize the matching error between two or more geometries is to insure that the matched components have the same area to periphery ratio. Therefore, the replication principle requires that all geometries have the same area-periphery ratio.

- Correct way to match the previous capacitors (the two C_2 capacitors are connected together):



- If $L_1 = L_2 = 2\mu\text{m}$, $W_2 = 2W_1 = 2\mu\text{m}$ and $\Delta x = 0.1\mu\text{m}$, the ratio of C_2 to C_1 can be written as,

$$\frac{C_2}{C_1} = \frac{2(2 - 0.2)(2 - 0.2)}{(2 - 0.2)(2 - 0.2)} = \frac{2 \cdot 1.8}{1.8} = 2 \rightarrow 0\% \text{ error in matching}$$

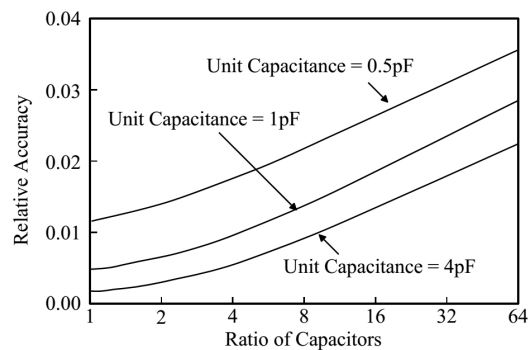
- The replication principle works for any geometry and includes transistors, resistors as well as capacitors.

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DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Replication Principle

- Capacitor Errors - Relative Accuracy
- Capacitor relative accuracy is proportional to the area of the capacitors and inversely proportional to the difference in values between the two capacitors.
- For example,

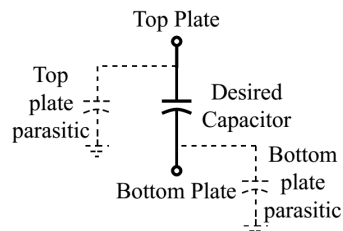


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DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Capacitor Errors - Parasitics

- Parasitics are normally from the top and bottom plate to AC ground which is typically the substrate.



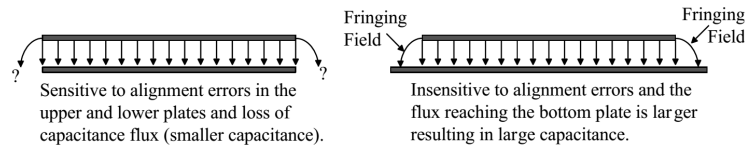
- Top plate parasitic is 0.01 to 0.001 of C_{desired}
- Bottom plate parasitic is 0.05 to 0.2 of C_{desired}

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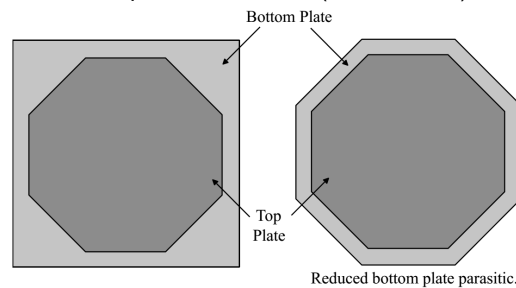
DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Layout Considerations on Capacitor Accuracy

- Decreasing Sensitivity to Edge Variation:



- A structure that minimizes the ratio of perimeter to area (circle is best).



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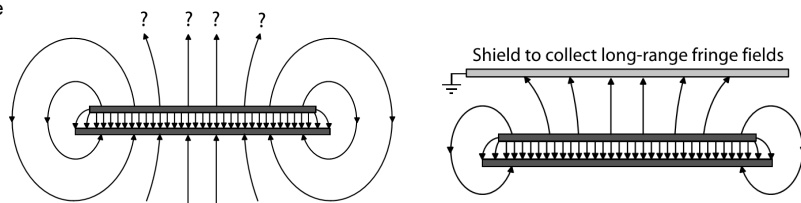
DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Accurate Matching of Capacitors†

- Accurate matching of capacitors depends on the following influence:

1. Mismatched perimeter ratios
2. Proximity effects in unit capacitor photolithography
3. Mismatched long-range fringe capacitance
4. Mismatched interconnect capacitance
5. Parasitic interconnect capacitance

- Long-range fringe capacitance:



- Obviously, there will be a tradeoff between matching and speed.

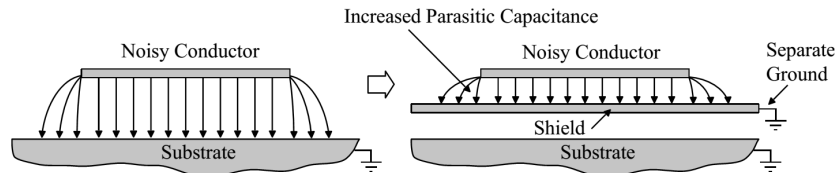
†M.J. McNutt, S. LeMarquis and J.L. Dunkley, "Systematic Capacitance Matching Errors and Corrective Layout Procedures," IEEE J. of Solid-State Circuit, vol. 29, No. 5, May 1994, pp. 611-616.

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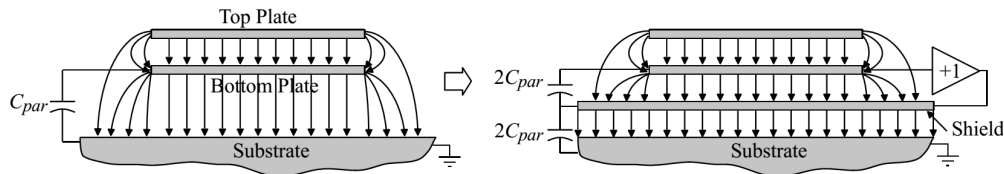
DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Shielding

- The key to shielding is to determine and control the electric fields.
- Consider the following noisy conductor and its influence on the substrate:



- Use of bootstrapping to reduce capacitor bottom plate parasitic:



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DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Definition of Temperature and Voltage Coefficients

- In general, a variable y which is a function of x , $y = f(x)$, can be expressed as a Taylor series,

$$y(x) \approx y(x_0) + a_1(x - x_0) + a_2(x - x_0)^2 + a_3(x - x_0)^3 + \dots$$

where the coefficients, a_i , are defined as,

$$a_1 = \left. \frac{df(x)}{dx} \right|_{x=x_0}, a_2 = \frac{1}{2} \left. \frac{d^2f(x)}{dx^2} \right|_{x=x_0}, \dots$$

- The coefficients, a_i , are called the first-order, second-order, temperature or voltage coefficients depending on whether x is temperature or voltage.
- Generally, only the first-order coefficients are of interest.
- In the characterization of temperature dependence, it is common practice to use a term called *fractional temperature coefficient*, TC_F , which is defined as,

$$TC_F(T=T_0) = \frac{1}{f(T_0)} \left. \frac{df(T)}{dT} \right|_{T=T_0} \text{ parts per million/}^\circ\text{C (ppm/}^\circ\text{C)}$$

- Or more simply as,

$$TC_F = \frac{1}{f(T)} \frac{df(T)}{dT} \text{ parts per million/}^\circ\text{C (ppm/}^\circ\text{C)}$$

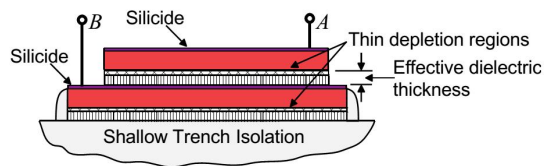
- A similar definition holds for fractional voltage coefficient.

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DEVIATION FROM IDEAL BEHAVIOR IN CAPACITORS

Capacitor Errors - Temperature and Voltage Dependence

- MOSFET Gate Capacitors:
 - Absolute accuracy $\approx \pm 10\%$
 - Relative accuracy $\approx \pm 0.2\%$
 - Temperature coefficient $\approx +25 \text{ ppm/C}^\circ$
 - Voltage coefficient $\approx -50 \text{ ppm/V}$
- Polysilicon-Oxide-Polysilicon Capacitors:
 - Absolute accuracy $\approx \pm 10\%$
 - Relative accuracy $\approx \pm 0.2\%$
 - Temperature coefficient $\approx +25 \text{ ppm/C}^\circ$
 - Voltage coefficient $\approx -20 \text{ ppm/V}$
- Metal-Dielectric-Metal Capacitors:
 - Absolute accuracy $\approx \pm 10\%$
 - Relative accuracy $\approx \pm 0.6\%$
 - Temperature coefficient $\approx +40 \text{ ppm/C}^\circ$
 - Voltage coefficient $\approx -20 \text{ ppm/V}$, 5 ppm/V^2
- Accuracies depend upon the size of the capacitors.

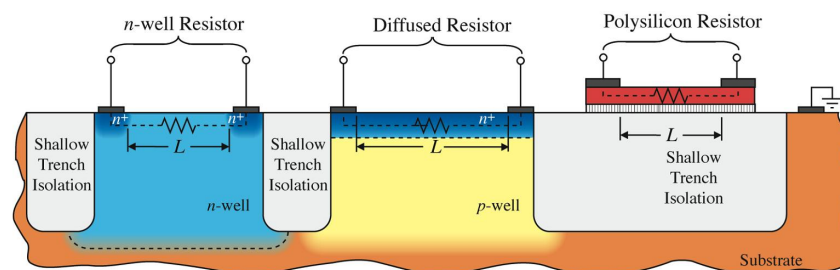


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RESISTORS

Types of Resistors Compatible with CMOS Technology

1. Diffused and/or implanted resistors.
2. Well resistors.
3. Polysilicon resistors.
4. Metal resistors.



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RESISTORS

Characterization of Resistors

1. Value

AC and DC resistance

2. Linearity

Does $V = IR$?

Velocity saturation of carriers

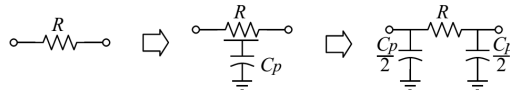
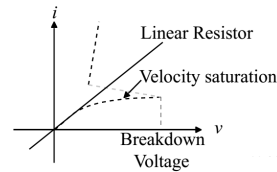
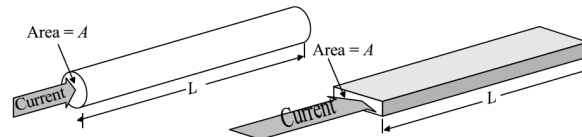
3. Power

$$P = VI = I^2R$$

4. Current

Electromigration

5. Parasitics



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RESISTORS

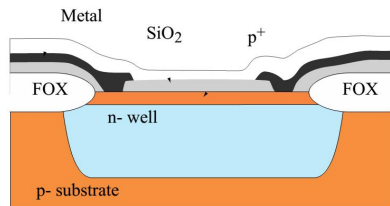
What is ppm?

- Ppm = Parts per million
- 1 ppm = $1/1\,000\,000 = 10^{-6}$
- Ppm is used for measuring concentration or density to extremely low volume or mass.
- Example:
 - Temperature coefficients $\approx +500 \text{ ppm}/^\circ\text{C}$ → Expressing accuracy of resistor when temperature changes
 - I.e. in 1°C , resistor may vary in $500/1.000.000 = 0.0005 \text{ Ohm}$

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RESISTORS

MOS Resistors - Source/Drain Resistor



Older LOCOS Technology

Diffusion:

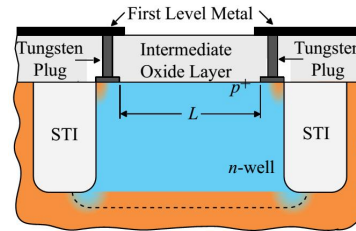
10-100 ohms/square

Absolute accuracy = $\pm 35\%$

Relative accuracy = 2% (5 μ m), 0.2% (50 μ m)

Temperature coefficient = +1500 ppm/ $^{\circ}$ C

Voltage coefficient $\approx$ 200 ppm/V



Ion Implanted:

500-2000 ohms/square

Absolute accuracy = $\pm 15\%$

Relative accuracy = 2% (5 μ m), 0.15% (50 μ m)

Temperature coefficient = +400 ppm/ $^{\circ}$ C

Voltage coefficient $\approx$ 800 ppm/V

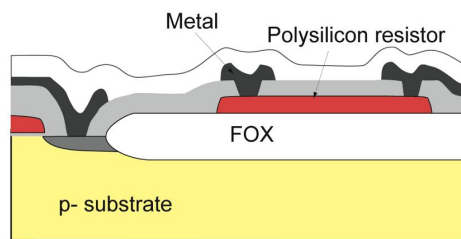
Comments:

- Parasitic capacitance to substrate is voltage dependent.
- Piezoresistance effects occur due to chip strain from mounting.

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RESISTORS

Polysilicon Resistor



Older LOCOS Technology

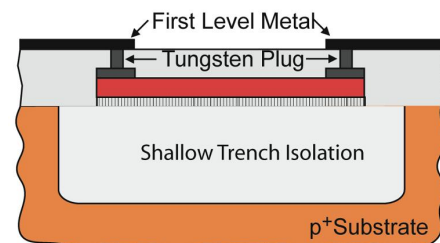
100-500 ohms/square (shielded)

Absolute accuracy = $\pm 3.0\%$

Relative accuracy = 2% (5 μ m)

Temperature coefficient = 500-1000 ppm/ $^{\circ}$ C

Voltage coefficient $\approx$ 100 ppm/V



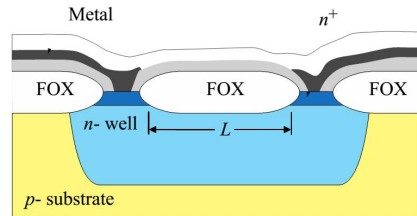
Comments:

- Used for fuzzes and laser trimming
- Good general resistor with low parasitics

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RESISTORS

N-well Resistor

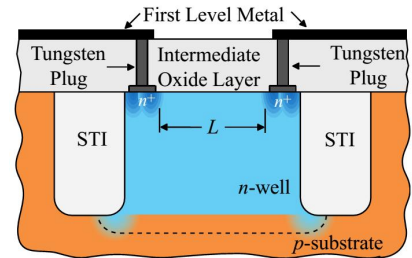


LOCOS Technology

1000-5000 ohms/square
 Absolute accuracy = $\pm 40\%$
 Relative accuracy $\approx 5\%$
 Temperature coefficient = 4000 ppm/ $^{\circ}\text{C}$
 Voltage coefficient is large ≈ 8000 ppm/V

Comments:

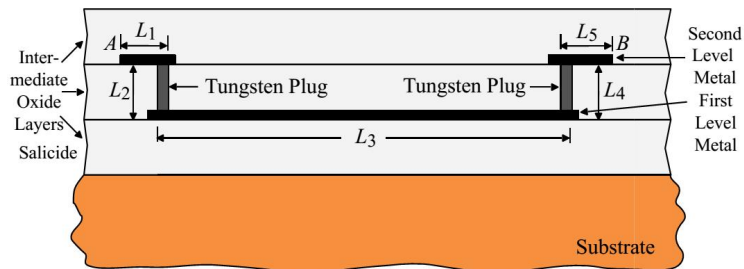
- Good when large values of resistance are needed.
- Parasitics are large and resistance is voltage dependent
- Could put a p+ diffusion into the well to form a pinched resistor



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RESISTORS

Metal as a Resistor



- Resistance from A to B = Resistance of segments L1, L2, L3, L4, and L5 with some correction subtracted because of corners.
- Sheet resistance:
 - 50-70 m Ω / $\square$ $\pm 30\%$ for lower or middle levels of metal
 - 30-40 m Ω / $\square$ $\pm 15\%$ for top level metal
- Watch out for the current limit for metal resistors.
- Contact resistance varies from 5 Ω to 10 Ω .
- Temperature coefficients $\approx +4000$ ppm/ $^{\circ}\text{C}$

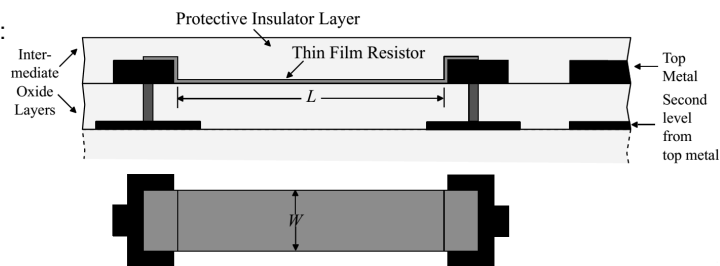
$T_j(^{\circ}\text{C})$	$T_r(^{\circ}\text{C})$	D_t
<85	85	1
100	85	0.63
110	85	0.48
125	85	0.32
150	85	0.18

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RESISTORS

Thin Film Resistors

- A high-quality resistor fabricated from a thin nickel-chromium alloy or a siliconchromium mixture.
- Uppermost metal layer:



- Performance:
 - Sheet resistivity is approximately 5-10 ohms/square
 - Temperature coefficients of less than 100 ppm/°C
 - Absolute tolerance of better than $\pm 0.1\%$ using laser trimming
 - Selectivity of the metal etch must be sufficient to ensure the integrity of the thinfilm resistor beneath the areas where metal is etched away.

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RESISTORS

Resistor Sizing

- Resistance value R is given by

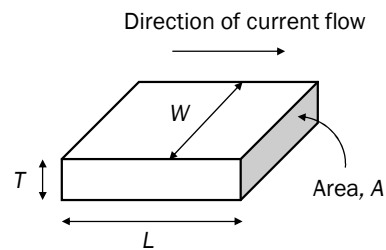
$$R = \frac{\rho L}{A} (\Omega)$$

$$R = \frac{\rho L}{WT} (\Omega)$$

- Where ρ_s is sheet resistivity [$\Omega/\square$] (ohm per square)

$$R = \left(\frac{\rho}{T}\right) \frac{L}{W} = \rho_s \frac{L}{W} (\Omega)$$

- From the layout point of view, a resistor has the value determined by the number of squares of resistance multiplied by ρ_s .



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RESISTORS

Example

- Given a polysilicon resistor like the picture with $W = 0.8\mu\text{m}$, $L = 20\mu\text{m}$. Calculate ρ_s , the number of squares of resistance, and the resistance value. Assume that ρ for polysilicon is $9 \times 10^{-4}\Omega\text{-cm}$ and polysilicon is $3000\text{\AA}$ thick. Ignore any contact resistance.

Solution:

- Calculate

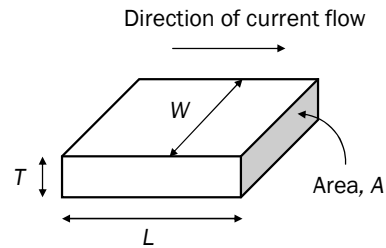
$$\rho_s = \left(\frac{\rho}{T}\right) = \frac{9 \times 10^{-4}\Omega\text{-cm}}{3000 \times 10^{-8}\text{cm}} = 30\Omega/\square$$

- The number of squares of resistance, N , is

$$N = \frac{L}{W} = \frac{20}{0.8} = 25$$

- The total resistance is

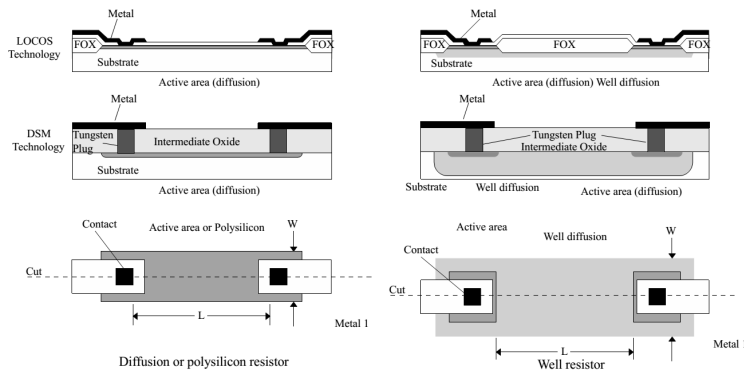
$$R = \rho_s \times N = 30 \times 25 = 750\Omega$$



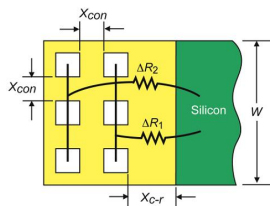
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RESISTORS

Resistor Layout Techniques



- End structure calculations:



$$\Delta R_1 = \frac{R_{\text{cont}}}{N_{\text{cont}}} + R_{\text{sh}}(\text{sil}) \left(\frac{X_{\text{c-r}} + 0.75 \cdot L_{\text{con}}}{W - 2 \cdot \Delta W_{\text{sil}}} \right)$$

(L_{con} = width of the contact)

$$\Delta R_2 = \frac{R_{\text{cont}}}{N_{\text{cont}}} + R_{\text{sh}}(\text{sil}) \left(\frac{X_{\text{c-r}} + X_{\text{con}} + 1.75 \cdot L_{\text{con}}}{W - 2 \cdot \Delta W_{\text{sil}}} \right)$$

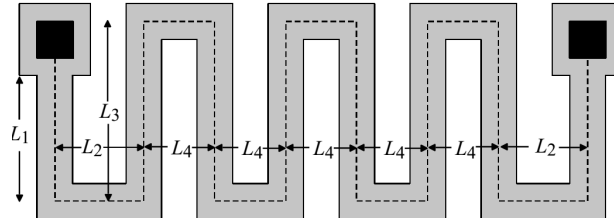
$$\Delta R_{\text{total}} = \left(\frac{1}{\Delta R_1} + \frac{1}{\Delta R_2} + \dots \right)^{-1}$$

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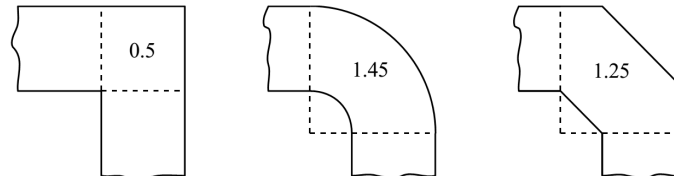
RESISTORS

Extending the Length of Resistors

- Snaked Resistors:



- Reduce the actual layout length.
- These resistors typically have model problems because of non-uniform current flow at the corners.
- Corner corrections:

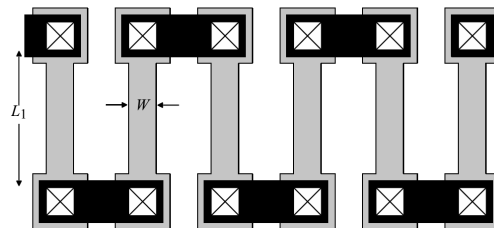


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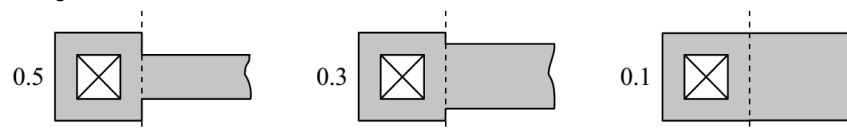
RESISTORS

Extending the Length of Resistors

- Link Resistors:



- For good matching between link resistors, keep the link length, L_1 , identical.
- Resistor Ending Influence:



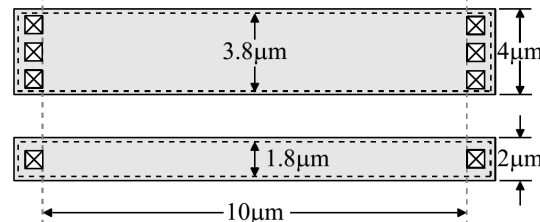
- Avoid "dogbone" resistors to minimize model errors.

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RESISTORS

Process Bias Influence on Resistors

- Process bias is where the dimensions of the fabricated geometries are not the same as the layout data base dimensions. Process biases introduce systematic errors.
- Consider the effect of over-etching. Assume that etching introduces a process bias of $0.1\mu\text{m}$. Two resistors designed to have a ratio of 2:1 have equal lengths but the widths are different by a factor of two.



- The actual matching ratio due to the etching bias is,

$$\frac{R_2}{R_1} = \frac{W_1}{W_2} = \frac{4-0.2}{2-0.2} = \frac{3.8}{1.8} = 2.11 \rightarrow 5.6\% \text{ error in matching}$$

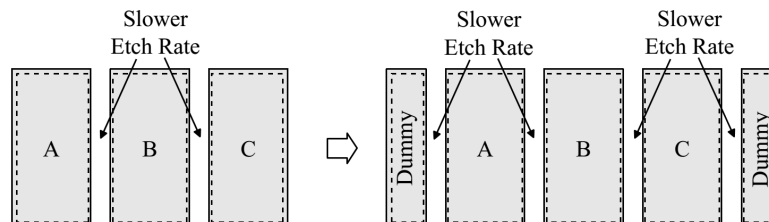
- Use the replication principle to eliminate this error.

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RESISTORS

Etch Rate Variations – Polysilicon Resistors

- The size of the area to be etched determines the etch rate. Smaller areas allow less access to the etchant while larger areas allow more access to the etchant. This is illustrated below:



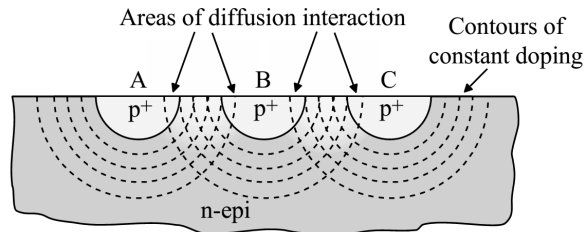
- The objective is to make $A = B = C$. In the left-hand case, B is larger due to the slower etch rates on both sides of B. In the right-hand case, the dummy strips have caused the etch rates on both sides of A, B and C to be identical leading to better matching.
- It may be advisable to connect the dummy strips to ground or some other low impedance node to avoid static electrical charge buildup.

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RESISTORS

Diffusion Interaction – Diffused Resistors

- Problem: Consider three adjacent p⁺ diffusions into a n-epitaxial region,



- If A, B, and C are resistors that are to be matched, we see that the effective concentration of B is larger than A or C because of diffusion interaction. This would cause the B resistor to be smaller even though the geometry is identical.
- Solution: Place identical dummy resistors to the left of A and right of C. Connect the dummy resistors to a low impedance to prevent the formation of floating diffusions that might increase the sensitivity to latchup.

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RESISTORS

Thermoelectric Effects

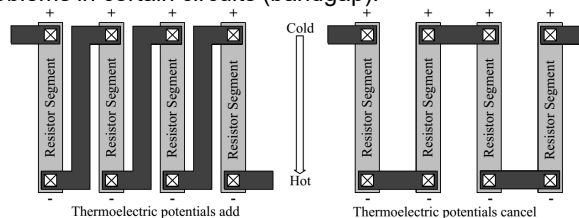
- The thermoelectric effect, also called the Seebeck effect, is a potential difference that is developed between two dissimilar materials that are at different temperatures. The potential developed is given as,

$$V_y = S \cdot \Delta T$$

where S = Seebeck coefficient ($\approx 0.4\text{mV}/^\circ\text{C}$)

ΔT = temperature difference between the two metals

- Thus, a temperature difference between the contacts to a resistor and the resistor of 1°C can generate a voltage of 0.4mV causing problems in certain circuits (bandgap).
- Two possible resistor layouts with regard to the thermoelectric effect:

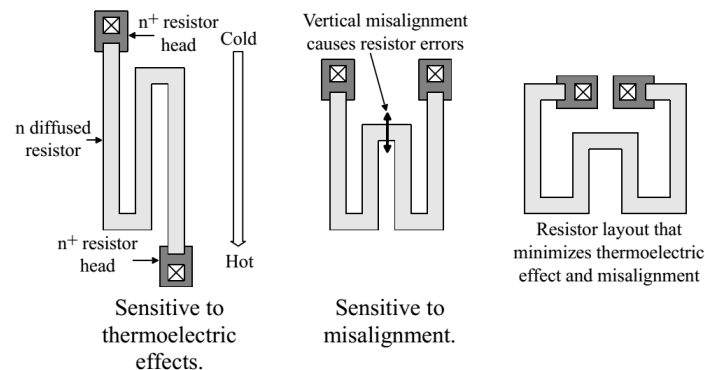


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RESISTORS

High Sheet Resistivity Resistor Layout

- High sheet resistivity resistors must use p+ or n+ in order to make contacts to metal.
- Thus, there is plenty of opportunity for the thermoelectric effect to cause problems if care is not taken. Below are three high sheet resistor layouts with differing thermoelectric performance.



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RESISTORS

MOS Passive RC Component Typical Performance Summary

Component Type	Range of Values	Absolute Accuracy	Relative Accuracy	Temperature Coefficient	Voltage Coefficient
MOSFET gate Cap.	6-7 fF/ μm^2	10%	0.1%	20ppm/ $^{\circ}\text{C}$	$\pm 20\text{ppm/V}$
Poly-Poly Capacitor	0.3-0.4 fF/ μm^2	20%	0.1%	25ppm/ $^{\circ}\text{C}$	$\pm 50\text{ppm/V}$
Metal-Metal Capacitor	0.1-1fF/ μm^2	10%	0.6%	-40ppm/ $^{\circ}\text{C}$	$\pm 1\text{ppm/V}$
Diffused Resistor	10-100 $\Omega/\text{sq.}$	35%	2%	1500ppm/ $^{\circ}\text{C}$	200ppm/V
Ion Implanted Resistor	0.5-2 k $\Omega/\text{sq.}$	15%	2%	400ppm/ $^{\circ}\text{C}$	800ppm/V
Poly Resistor	30-200 $\Omega/\text{sq.}$	30%	2%	1500ppm/ $^{\circ}\text{C}$	100ppm/V
n-well Resistor	1-10 k $\Omega/\text{sq.}$	40%	5%	8000ppm/ $^{\circ}\text{C}$	10kppm/V
Top Metal Resistor	30 m $\Omega/\text{sq.}$	15%	2%	4000ppm/ $^{\circ}\text{C}$	-
Lower Metal Resistor	70 m $\Omega/\text{sq.}$	28%	3%	4000ppm/ $^{\circ}\text{C}$	-

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INDUCTORS

Characterization of Inductors

1. Value of the inductor

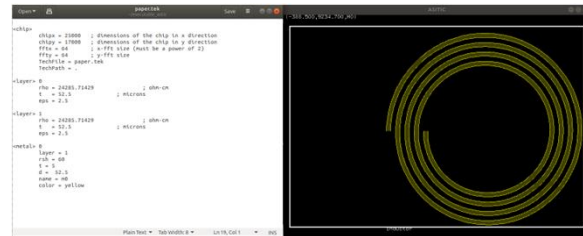
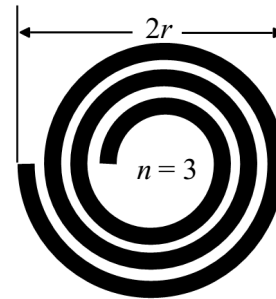
Spiral inductor†:

$$L \approx \mu_0 n^2 r = 4\pi \times 10^{-7} n^2 r \approx 1.2 \times 10^{-6} n^2 r$$

2. Quality factor, $Q = \frac{\omega L}{R}$

3. Self-resonant frequency: $f_{shelf} = \frac{1}{\sqrt{LC}}$

4. Parasitic and inter-winding capacitances



† H.M. Greenhouse, "Design of Planar Rectangular Microelectronic Inductors," IEEE Trans. Parts, Hybrids, and Packaging, vol. 10, no. 2, June 1974, pp. 101-109.

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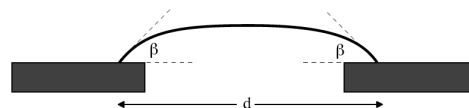
INDUCTORS

Candidates for inductors in CMOS technology:

- Candidates for inductors in CMOS technology are:

1. Bond wires
2. Spiral inductors
3. Multi-level spiral
4. Solenoid

- Bond wire Inductors:

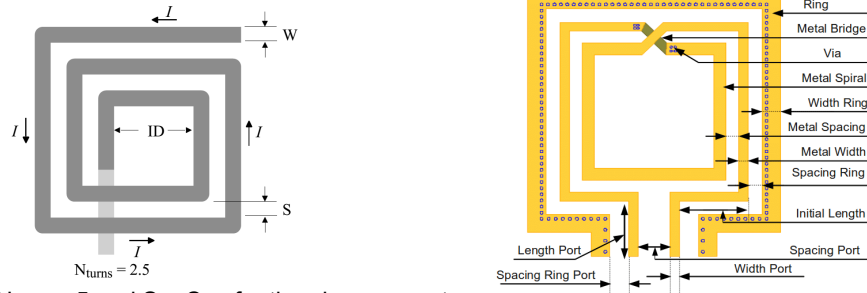


- Function of the pad distance d and the bond angle beta
- Typical value is 1nH/mm which gives 2nH to 5nH in typical packages
- Series loss is 0.2Ω/mm for 1 mil diameter aluminum wire

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INDUCTORS

Planar Spiral Inductors in CMOS Technology

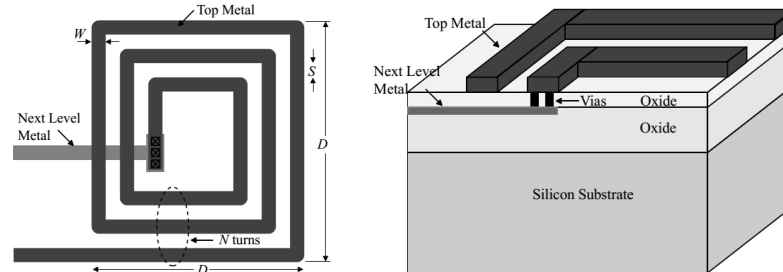


- Typically: $3 < N_{\text{turns}} < 5$ and $S = S_{\text{min}}$ for the given current
- Select the ID , N_{turns} , and W so that ID allows sufficient magnetic flux to flow through the center.
- Loss Mechanisms:
 - Skin effect
 - Capacitive substrate losses
 - Eddy currents in the silicon

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INDUCTORS

Planar Spiral Inductors on a Lossy Substrate

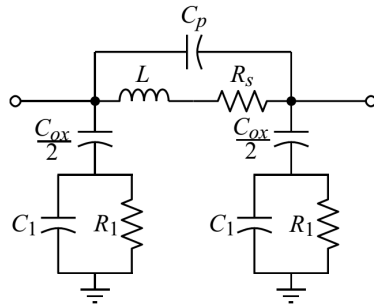


- Spiral inductor is implemented using metal layers in CMOS technology
- Topmost metal is preferred because of its lower resistivity
- More than one metal layer can be connected together to reduce resistance or area
- Accurate analysis of a spiral inductor requires complex electromagnetic simulation
- Optimize the values of W , S , and N to get the desired L , a high Q , and a high self-resonant frequency
- Typical values are $L = 1\text{-}8\text{ nH}$ and $Q = 3\text{-}6$ at 2 GHz

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INDUCTORS

Inductor Modeling



where

$\mu_0 = 4\pi \times 10^{-7}$ H/m (vacuum permeability)

σ = conductivity of the metal

a = distance from the center of the inductor to the middle of the windings

L = total length of the spiral

t = thickness of the metal

δ = skin depth given by $\delta = \sqrt{2/W\mu\sigma}$

$G_{\text{sub}}(C_{\text{sub}})$ is a process-dependent parameter

$$L \approx \frac{37.5\mu_0 N^2 a^2}{11D-14a}$$

$$C_{ox} = W \cdot L \cdot \frac{\epsilon_{ox}}{t_{ox}}$$

$$R_s \approx \frac{L}{W\sigma\delta(1-e^{-t/\delta})} \text{ (low freq. resistive loss)}$$

$$R_1 \approx \frac{WLC_{\text{sub}}}{2} \text{ (eddy current substrate loss)}$$

$$C_p = NW^2L \cdot \frac{\epsilon_{ox}}{t_{ox}} \text{ (overlap and coupling)}$$

$$C_1 \approx \frac{2}{WLC_{\text{sub}}} \text{ (substrate capacitance)}$$

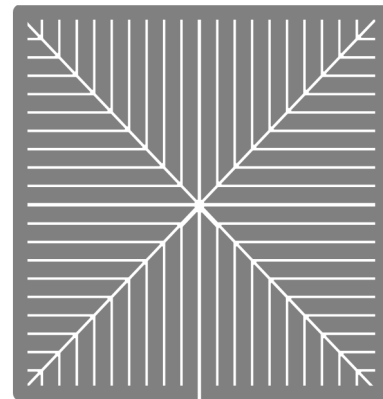
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INDUCTORS

Reduction of Capacitance to Ground

Comments concerning implementation:

1. Put a metal ground shield between the inductor and the silicon to reduce the capacitance.
 - Should be patterned so flux goes through but electric field is grounded
 - Metal strips should be orthogonal to the spiral to avoid induced loop current
 - The resistance of the shield should be low to terminate the electric field
2. Avoid contact resistance wherever possible to keep the series resistance low.
3. Use the metal with the lowest resistance and farthest away from the substrate.
4. Parallel metal strips if other metal levels are available to reduce the resistance.



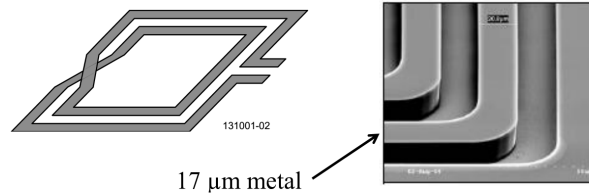
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INDUCTORS

Inductor Improvements

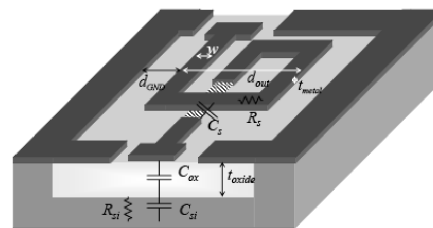
Symmetrical Layout:

- Good for differential circuits
- Higher Q
- Can achieve a center tap



Q Improvement:

- Substrate replaced with trenched silicon islands[†]



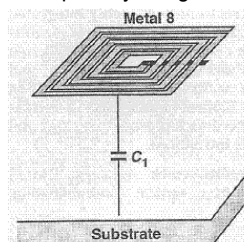
[†] M. Raieszadeh, Integrated Inductors on Trenched Silicon Islands, MS Thesis, School of Electrical and Computer Engineering, Georgia Institute of Technology, April 2005

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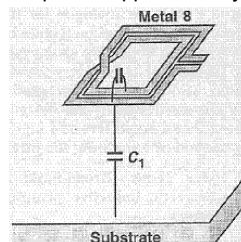
INDUCTORS

Multi-Level Spiral Inductors

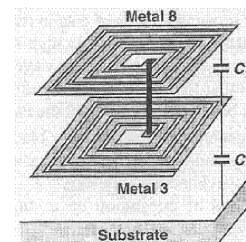
- Use of more than one level of metal to make the inductor.
 - Can get more inductance per area
 - Can increase the interwire capacitance so the different levels are often offset to get minimum overlap.
 - Multi-level spiral inductors suffer from contact resistance (must have many parallel contacts to reduce the contact resistance).
 - Metal especially designed for inductors is top level approximately 4μm thick.



$Q = 5-6$, $f_{SR} = 20-40\text{GHz}$



$Q = 10-11$, $f_{SR} = 15-30\text{GHz}$ ¹



Good for high L in small area

¹ The skin effect and substrate loss appear to be the limiting factor at higher frequencies of self-resonance.

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INDUCTORS

Summary of Inductors

- Scaling? To reduce the size of the inductor would require increasing the flux density which is determined by the material the flux flows through. Since this material will not change much with scaling, the inductor size will remain constant.
- Increase in the number of metal layers will offer more flexibility for inductor and transformer implementation.
- Performance:
 - Inductors
 - Limited to nanohenrys
 - Very low Q factor (3-5)
 - Transformers
 - Reasonably easy to build and work well using stacked inductors

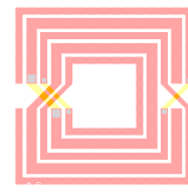
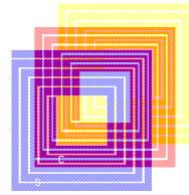
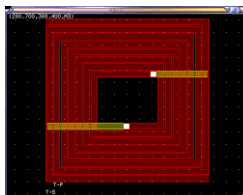
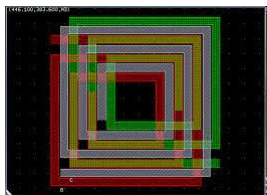
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INDUCTORS

Summary of Inductors

- ASITIC[†] – A CAD tool that aids the RF circuit designer to optimize and model spiral inductors, transformers, capacitors, and substrate coupling. ASITIC calculations include the electrically induced losses and coupling as well as the magnetically induced eddy current losses. Skin effect and proximity effects, or eddy currents in the metallization, are also included.

[†] <http://rfic.eecs.berkeley.edu/~niknejad/asitic.html>

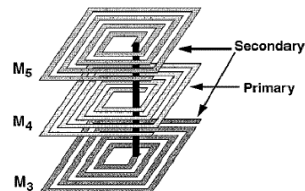


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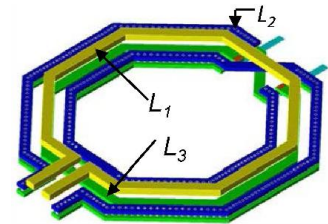
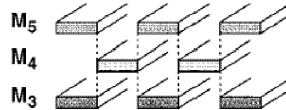
INDUCTORS

Transformers

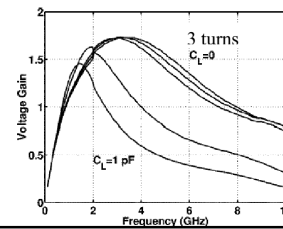
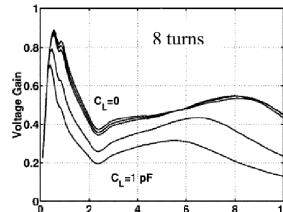
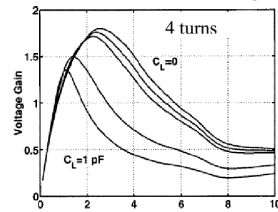
- Transformer structures are easily obtained using stacked inductors as shown below for a 1:2 transformer.



Method of reducing the inter-winding capacitances.



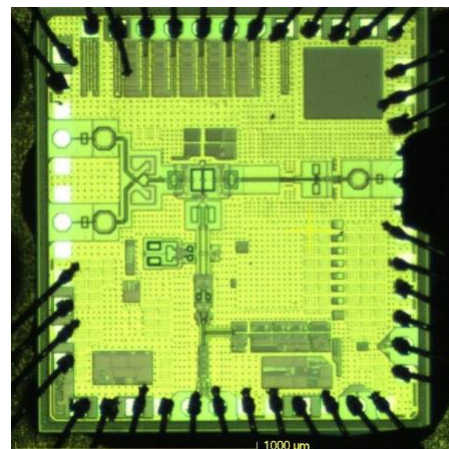
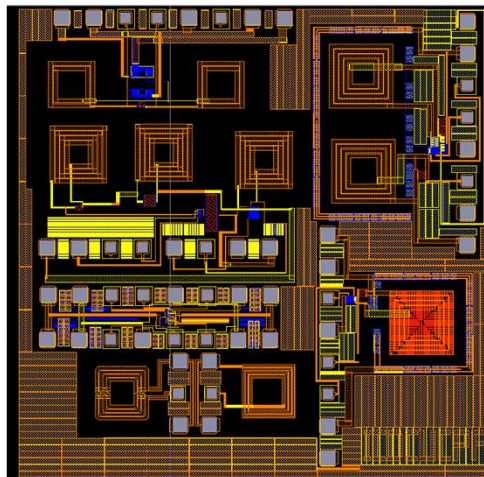
Measured 1:2 transformer voltage gains:



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LAYOUT OF RESISTOR, CAPACITOR, INDUCTORS

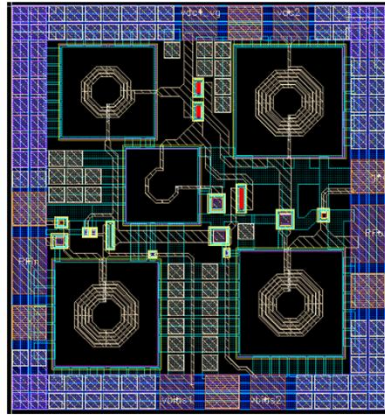
- Passive components are mainly used in analog circuits and RF ICs



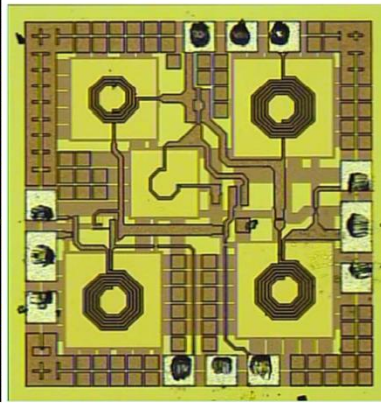
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LAYOUT OF RESISTOR, CAPACITOR, INDUCTORS

- Passive components are mainly used in analog circuits and RF ICs



(a)



(b)

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**THE END
(THIS PART)**

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